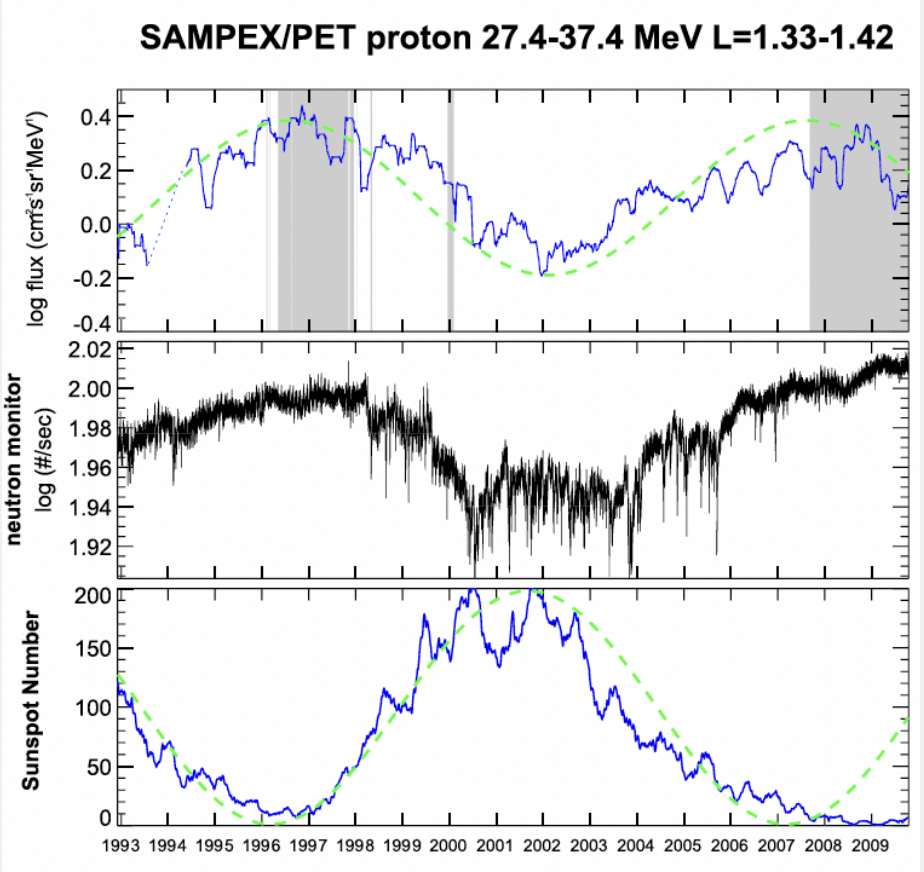
University of Colorado **Boulder**

Radiation Belt Dynamics + Sources + Losses (Continued)

Xinlin Li

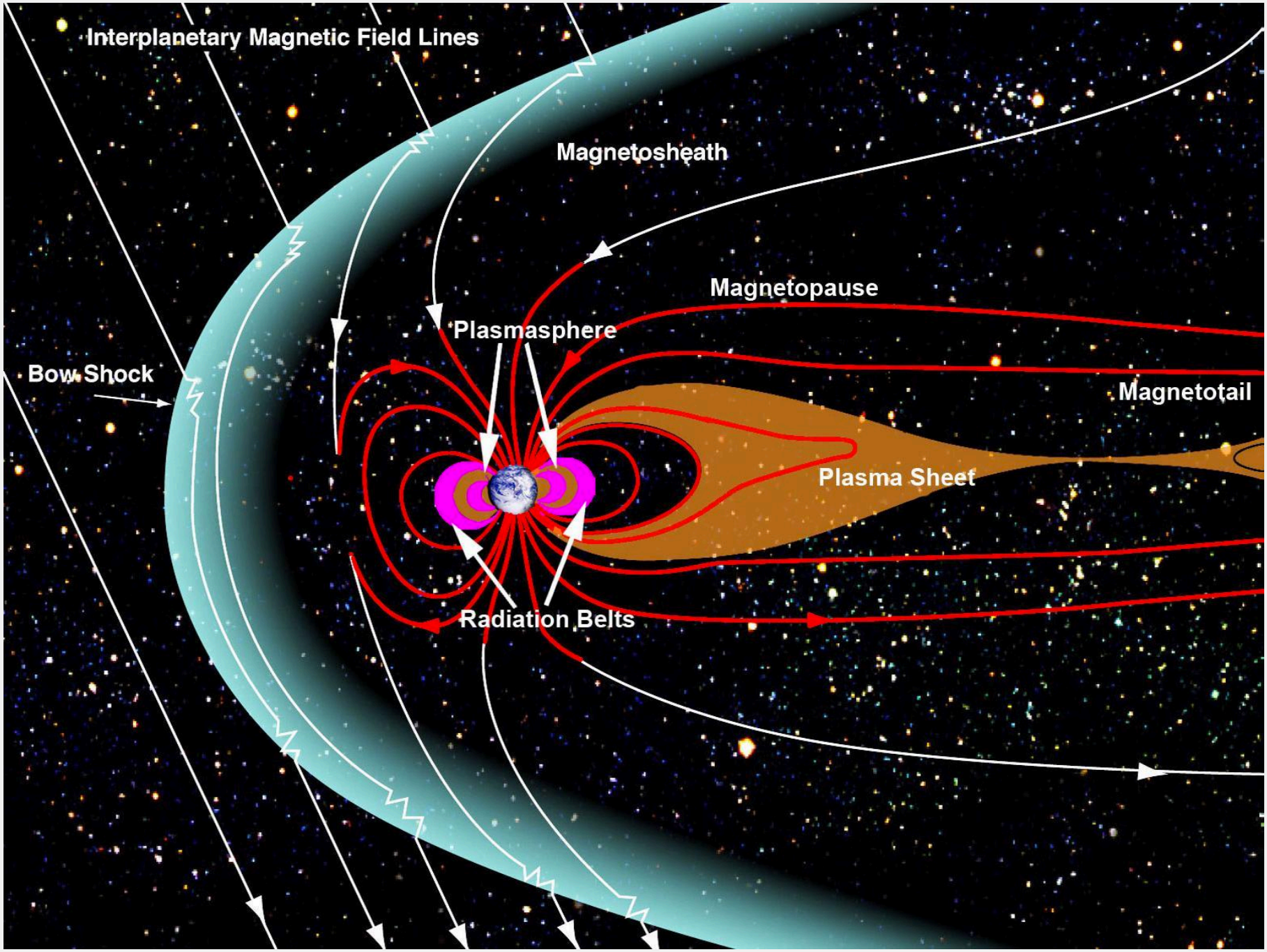
Radiation Belt Loss: Inner Belt

- Inner belt protons are lost due to charge exchange with H atoms in atmosphere
  - At first, **H^+ is fast** and **H is slow**
  - **H** gives electron to **H^+**; now **H^+ is slow** and **H is fast**
  - **H** escapes because it is not tied to B-field



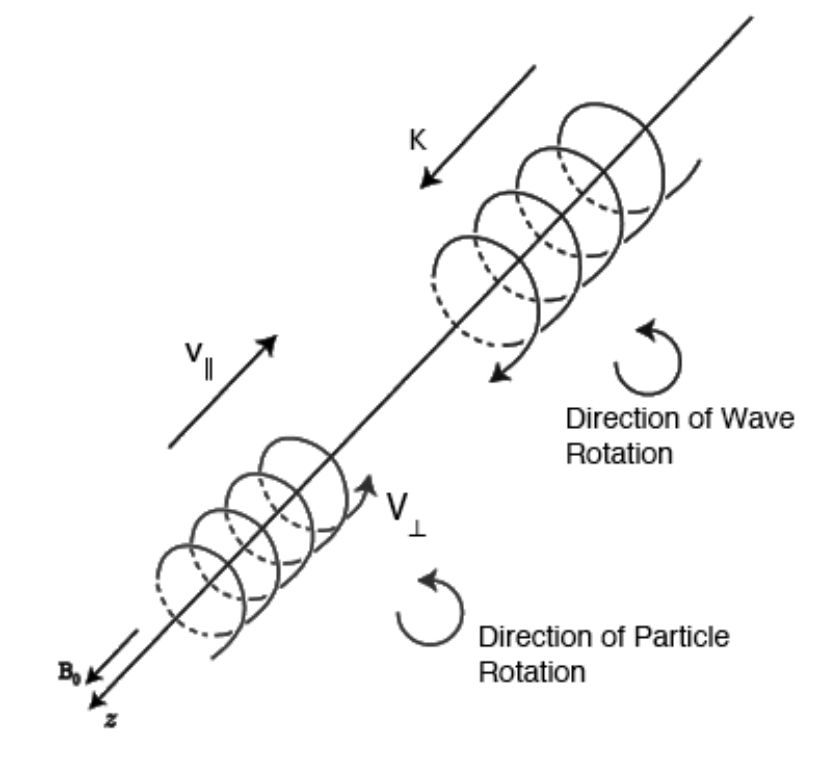
(Li et al., JGR, 2020)

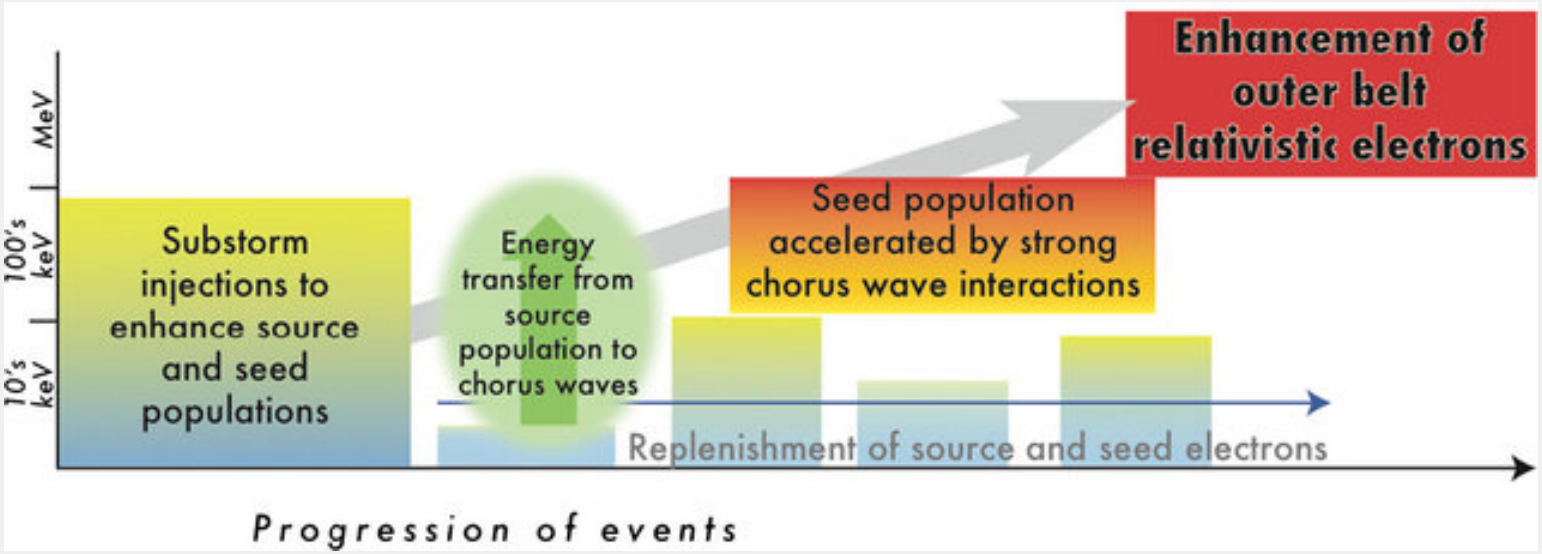
Radiation Belt Sources: Outer Belt



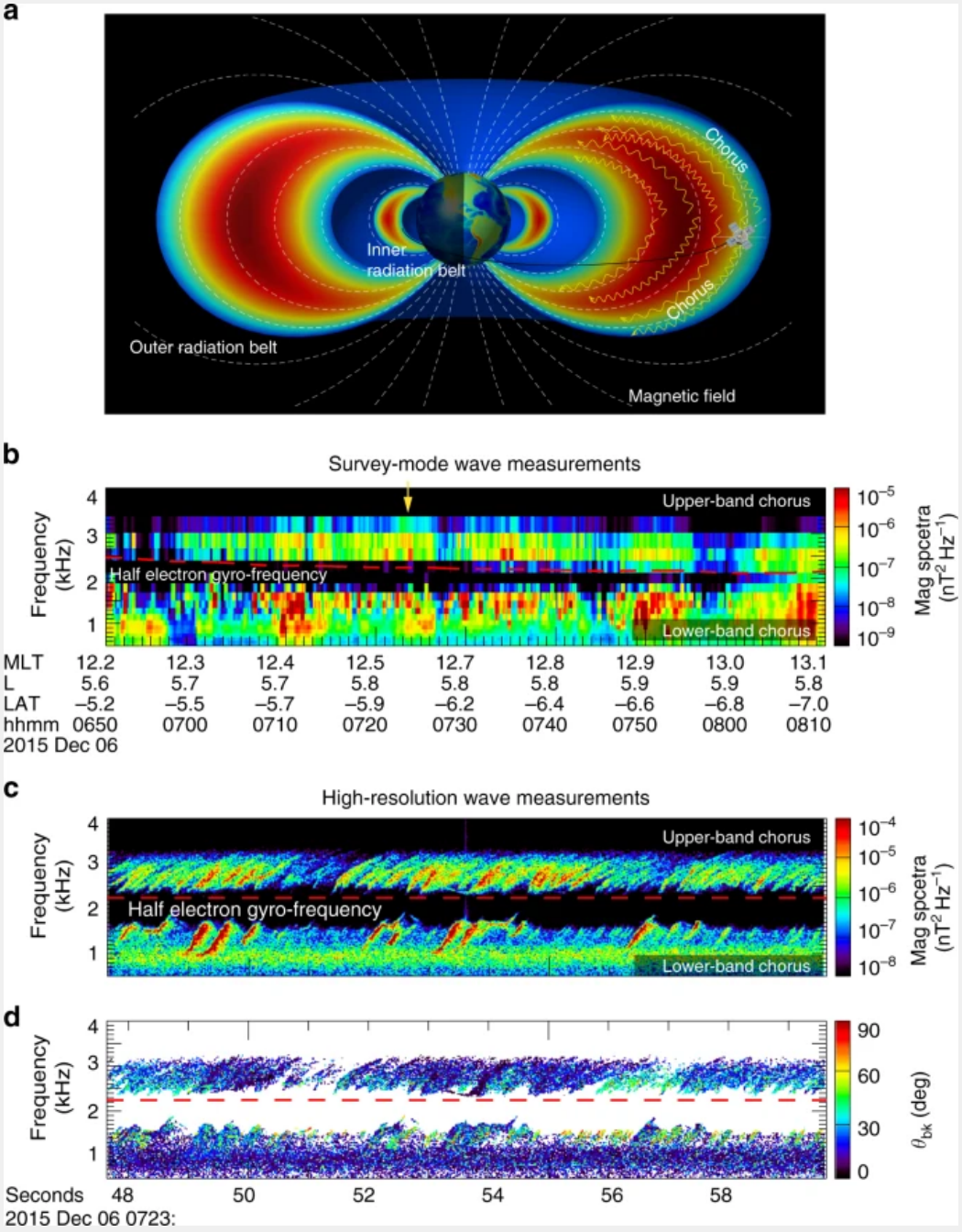
Outer Belt Energization

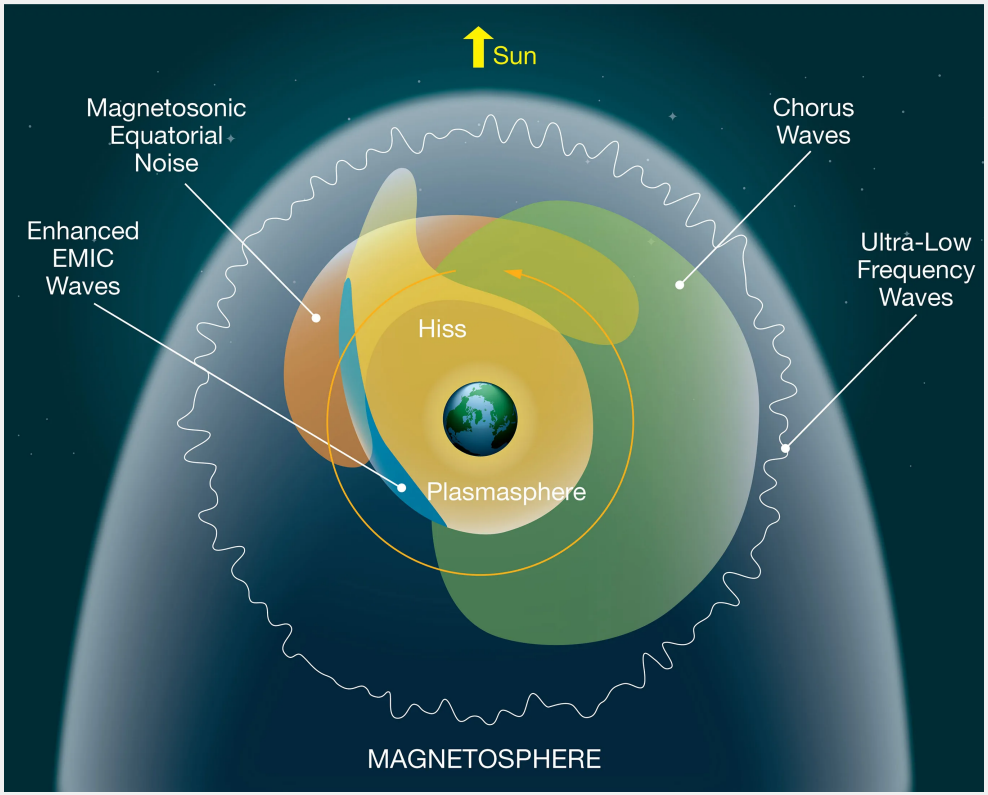
- Now we have plasma sheet electrons (10–250 keV) transported into the outer belt region (3 < L < 6):
  "seed" electrons
- Energized to MeV energies by interaction with electromagnetic waves ("chorus")
  - **"cyclotron resonance"**

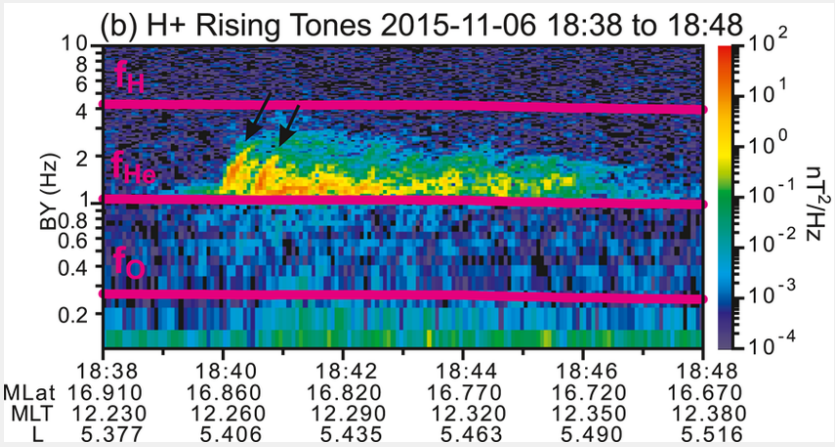




Plasma Waves and their Distributions

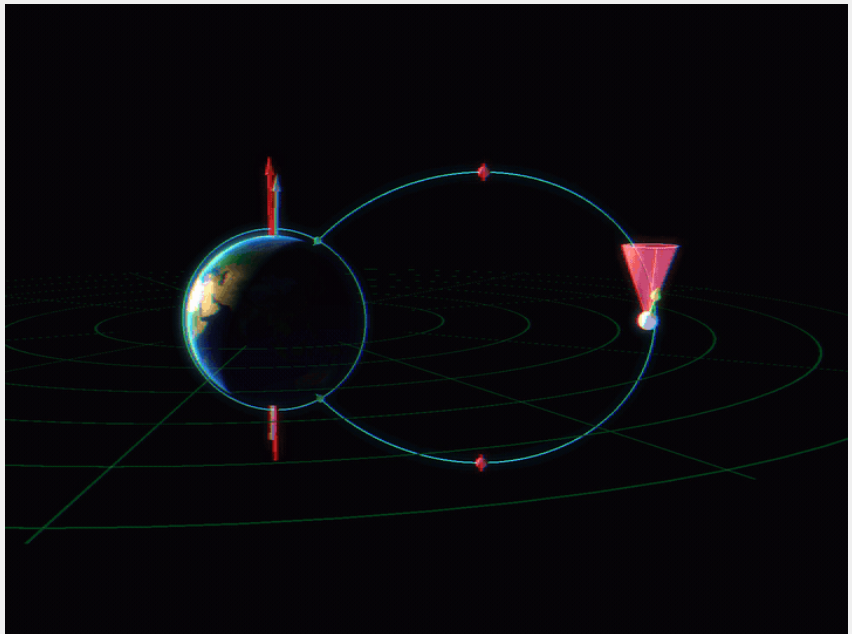


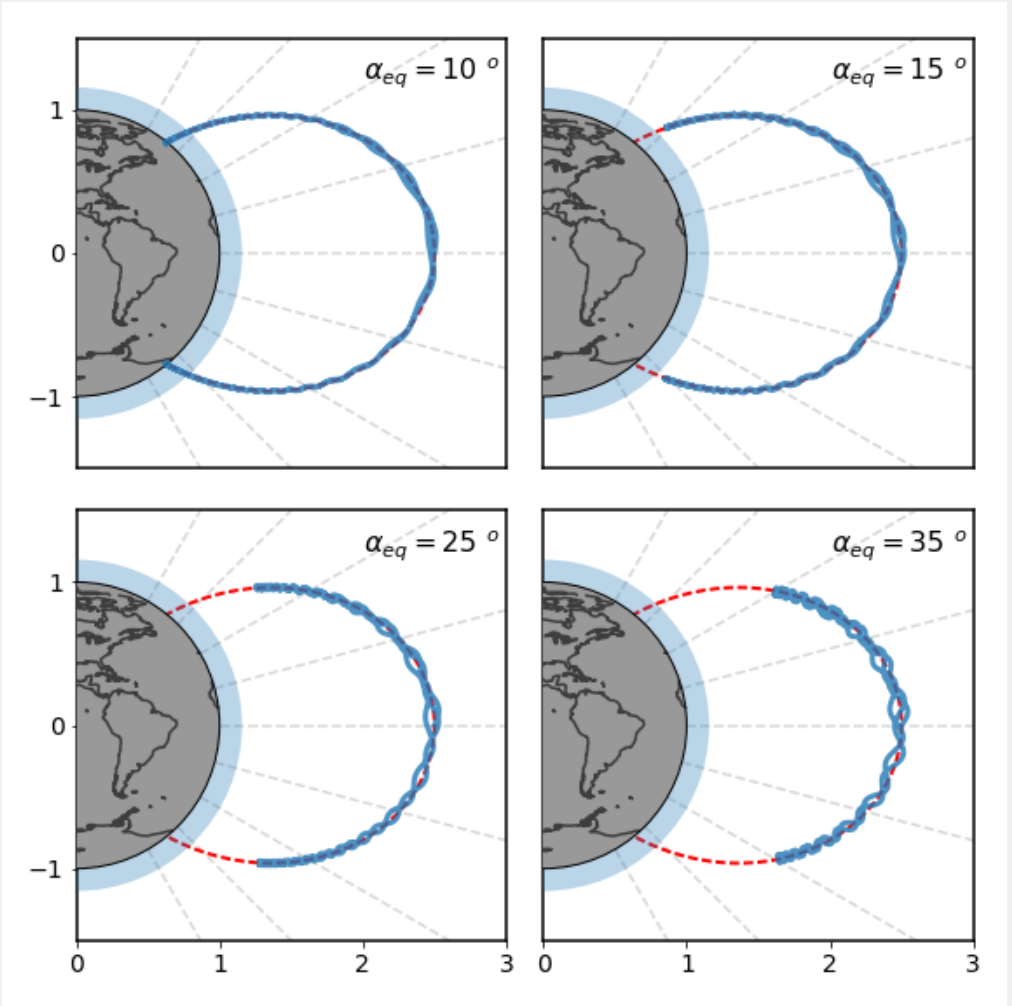




Radiation Belt Losses: Precipitation

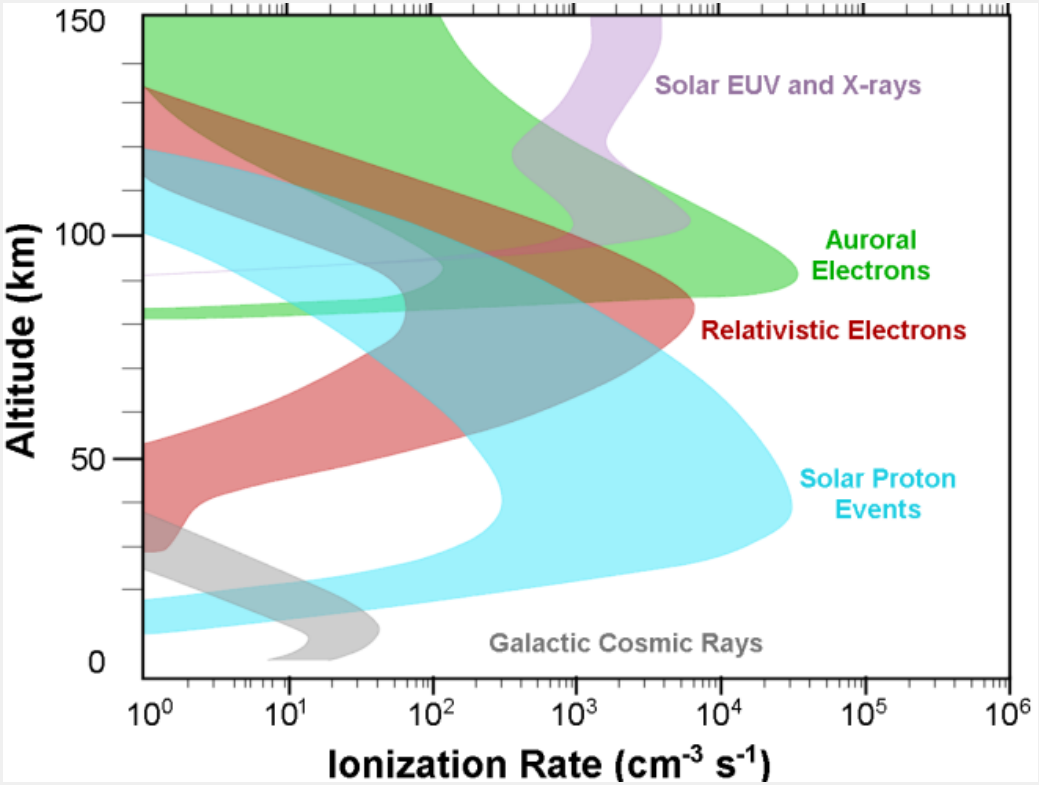
- Concept: **The Loss Cone**
  - The equatorial pitch angle α_{eq} of particles determines where they will mirror
  - α_{eq} = 90: mirroring at equator
  - α_{eq} = 45: mirroring at higher latitude
  - α_{eq} = 0: will hit the Earth!
- Loss cone angle α_{LC} – particles with lower pitch angle will reach 100 km altitude and be lost the atmosphere
- We call this **radiation belt precipitation**

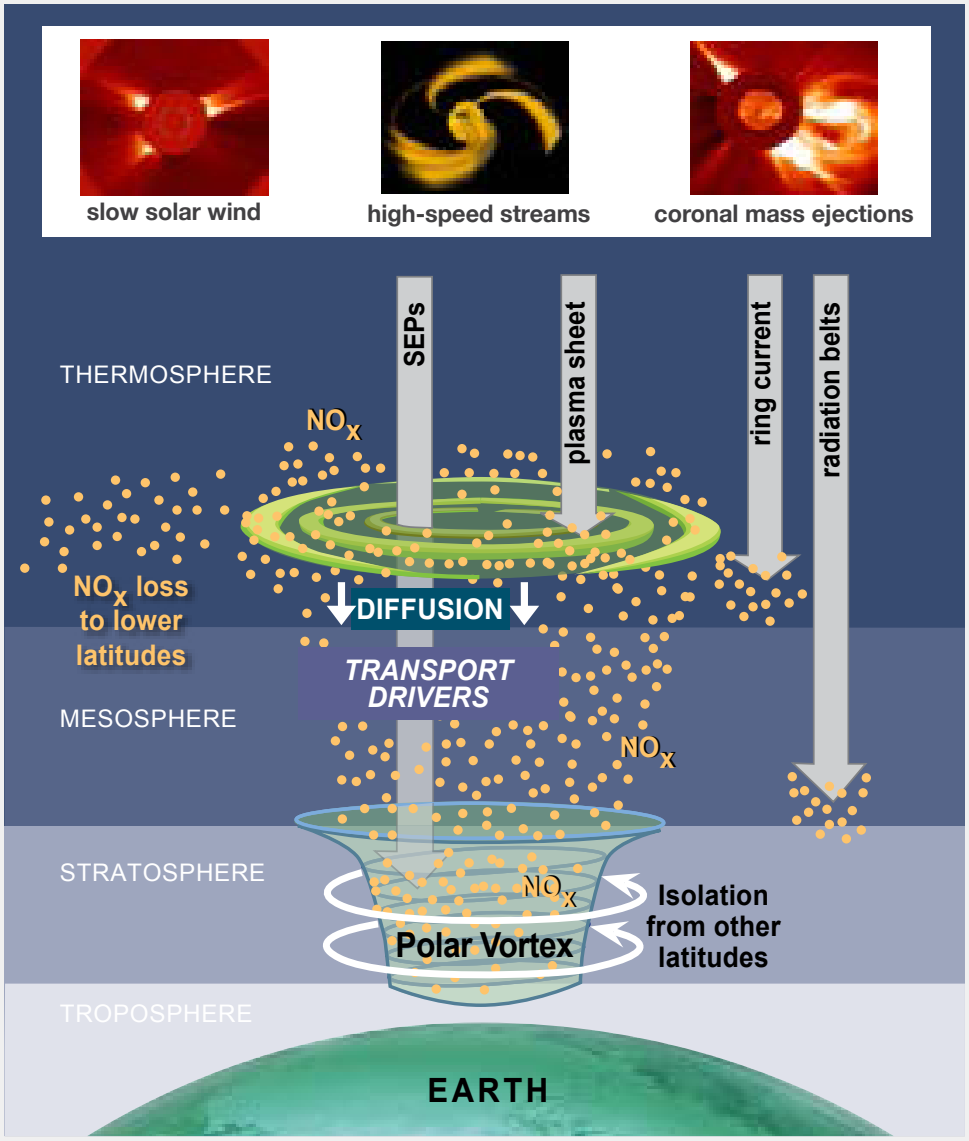




Atmospheric Effects of Precipitation

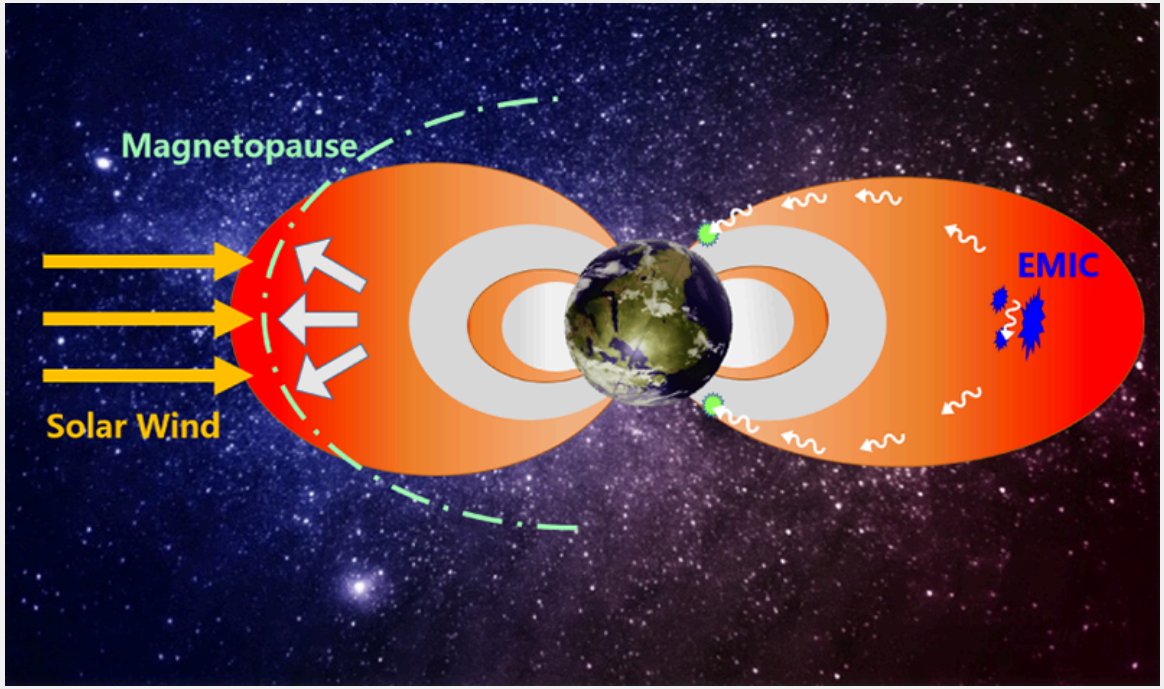
- Ionization in the upper atmosphere (temporary; recombines quickly)
- Chemical effects: NOx, HOx enhancement, ozone destruction

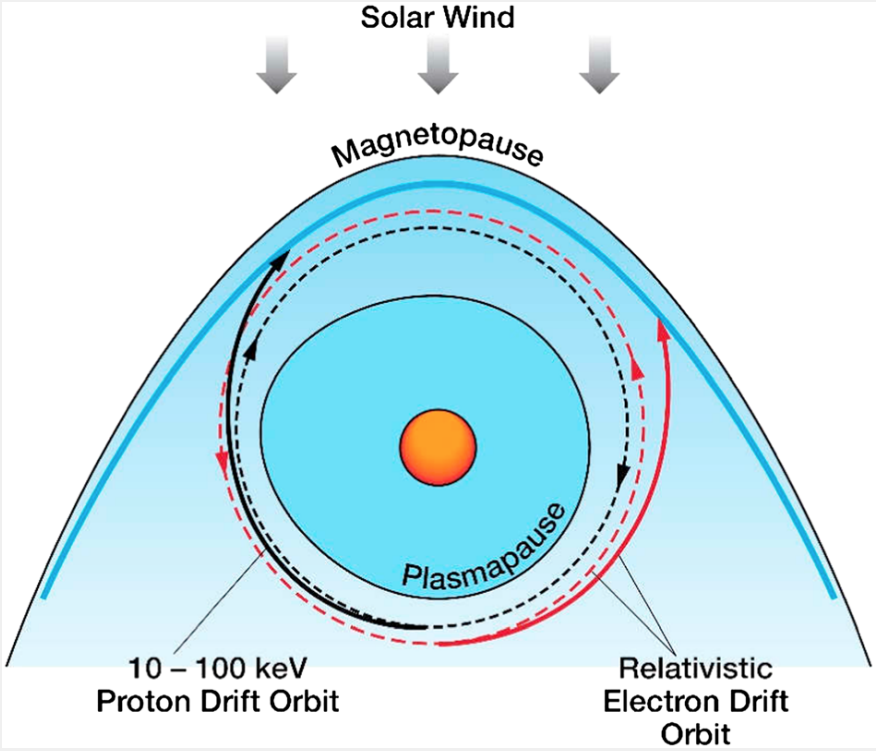




Radiation Belt Losses: Magnetopause Shadowing

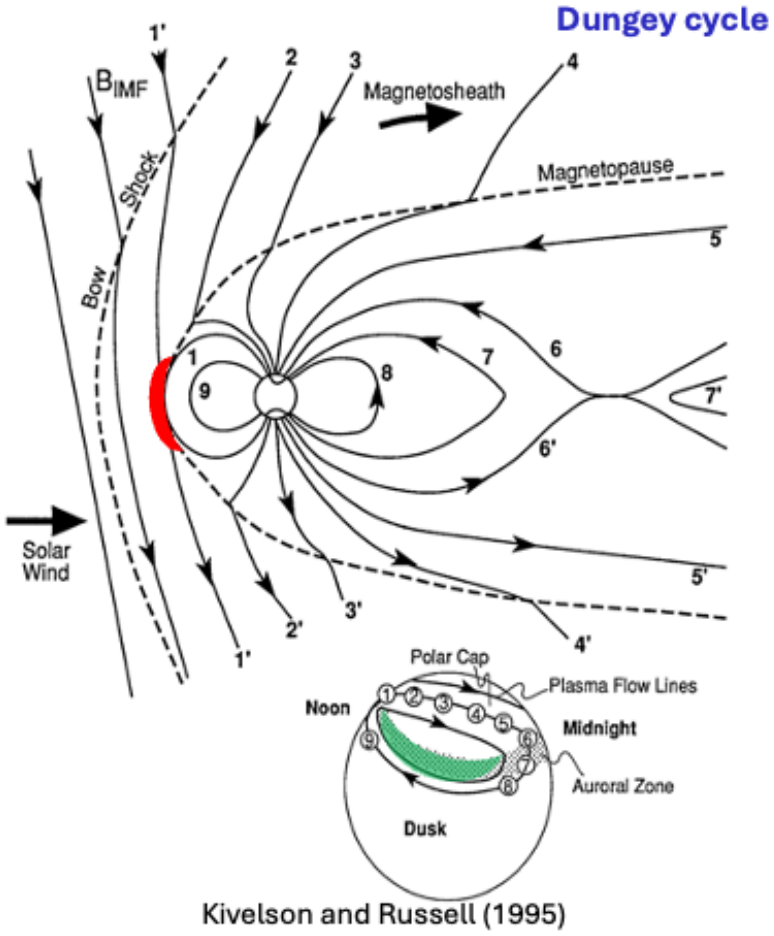
- Particles at outer edge of belt on nightside drift around to dayside; during compressed magnetosphere, outer belt may extend outside magnetopause
- Once particles reach magnetopause, they can escape on open field lines



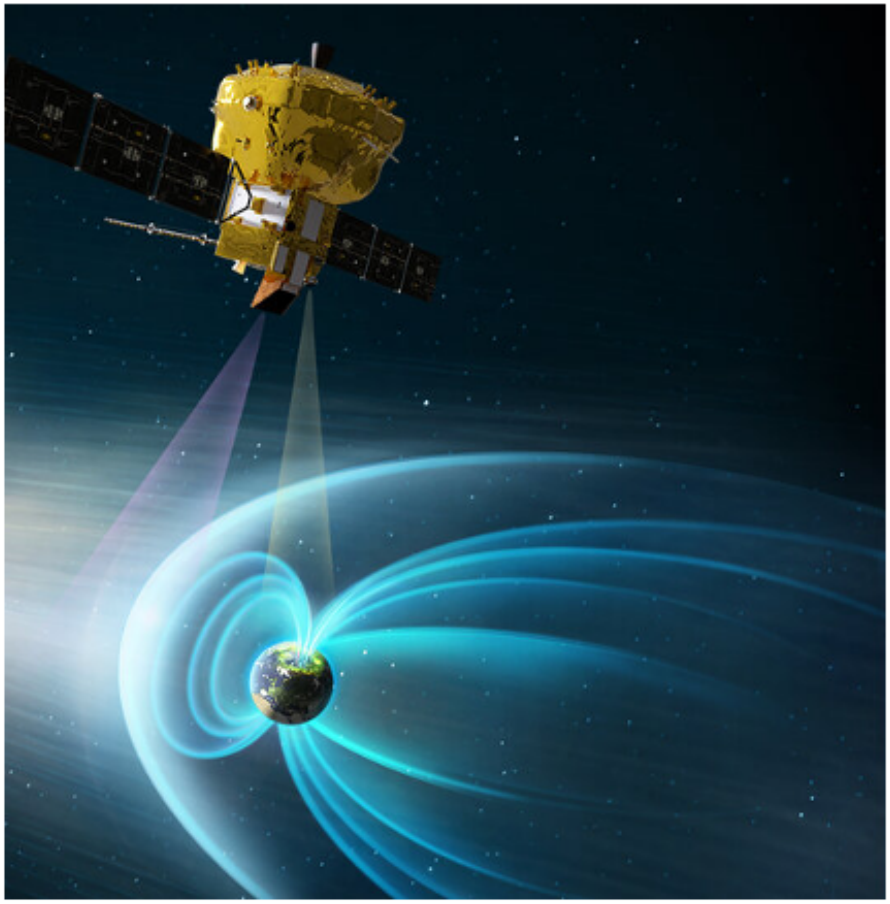


Observing Magnetopause Location

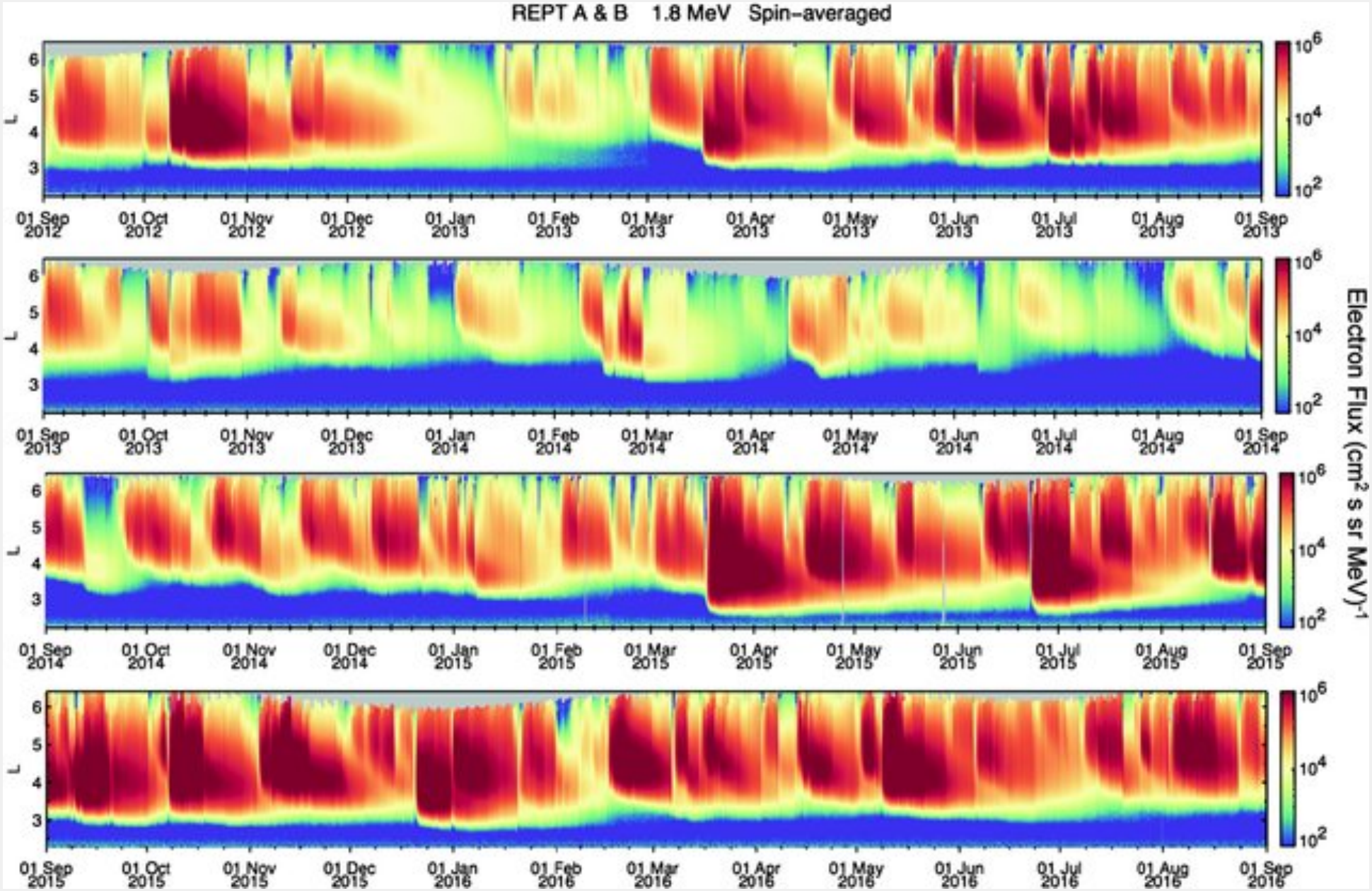
SMILE, a joint ESA–Chinese Academy of Sciences mission, was launched in May 2026 and began science operations in September 2026. Its ultraviolet imager has successfully obtained global images of the northern auroral oval, while its soft X-ray imager is being optimized to provide global observations of the dayside magnetopause and polar cusps.



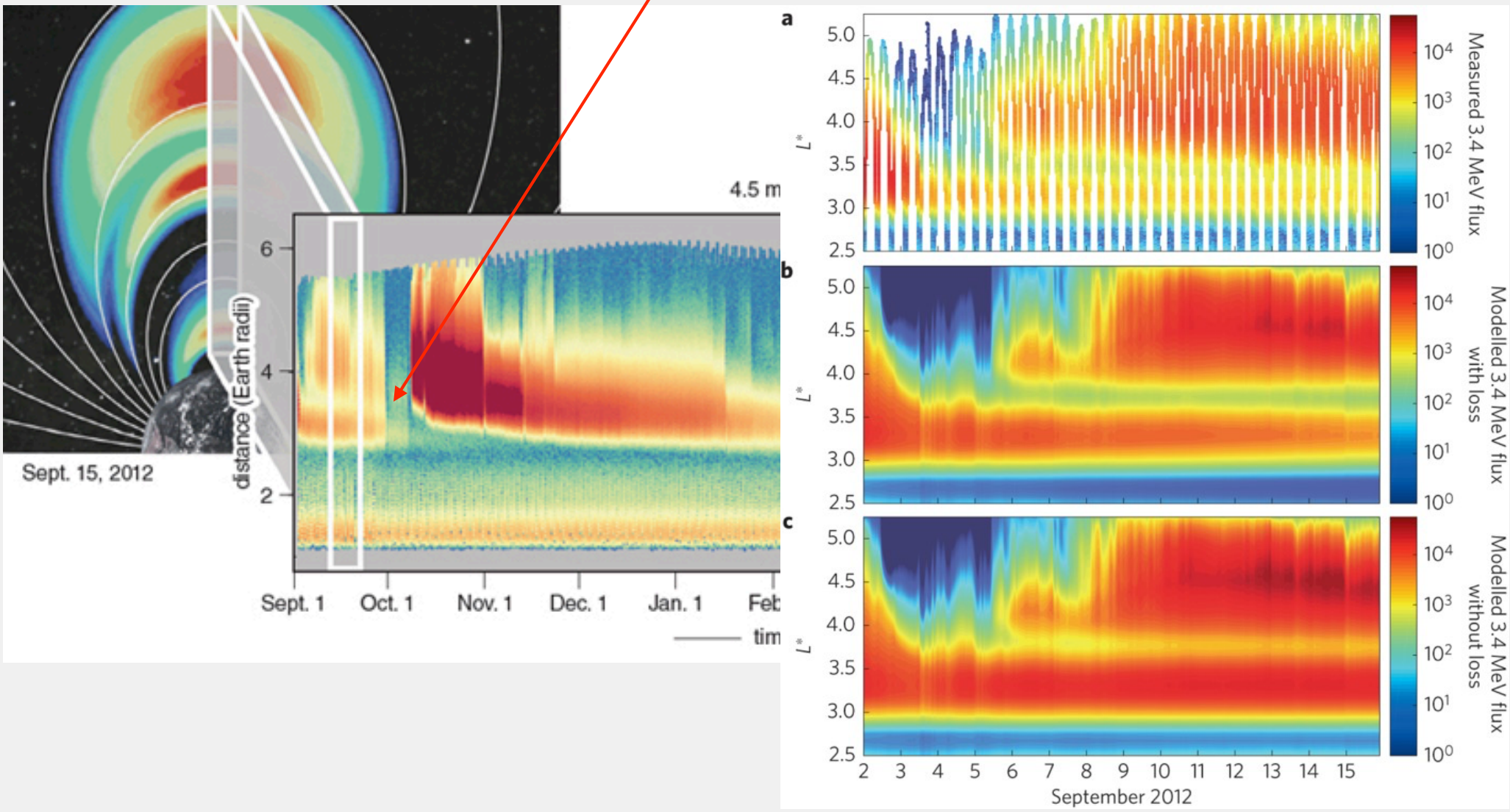
Kivelson and Russell (1995)


Picture credit: ESA

Radiation Belt Variability (drop out/decrease and enhancements)

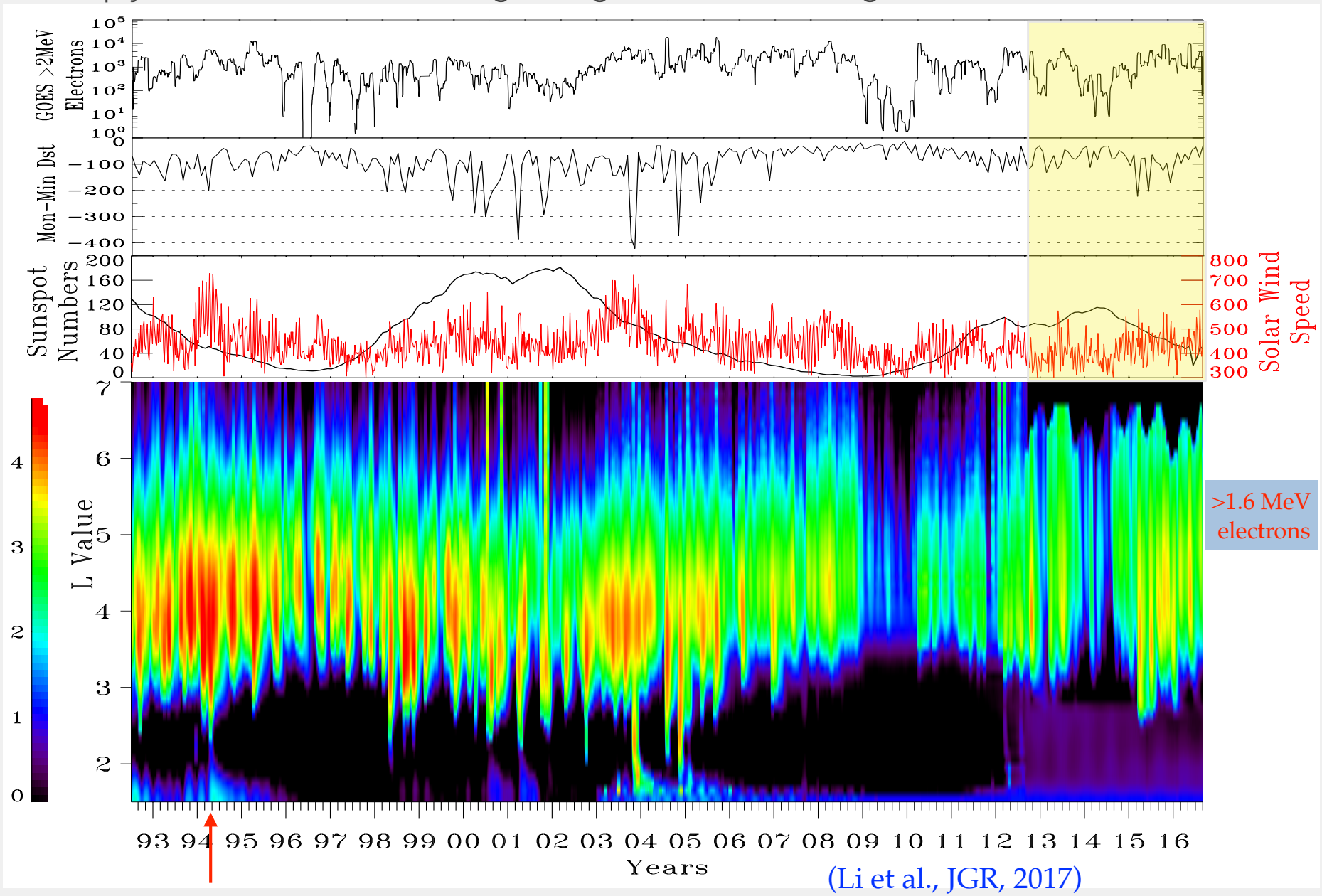


Zoomed in View of a Dropout



Solar Cycle Variation

- Solar Cycle is not the dominant driver of radiation belt variability, but there is some correlation
  - Simply because there are more geomagnetic storms during solar maximum and soon after

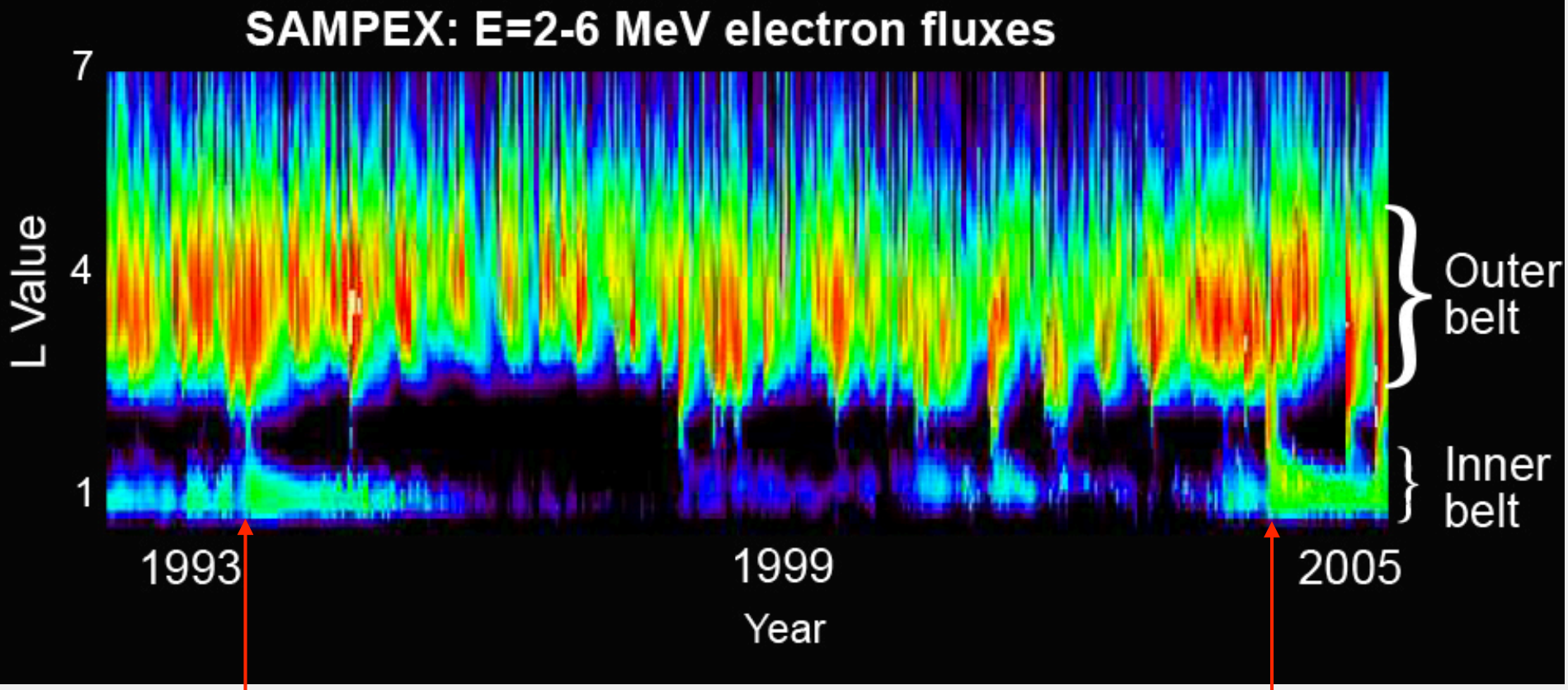


(Li et al., JGR, 2017)

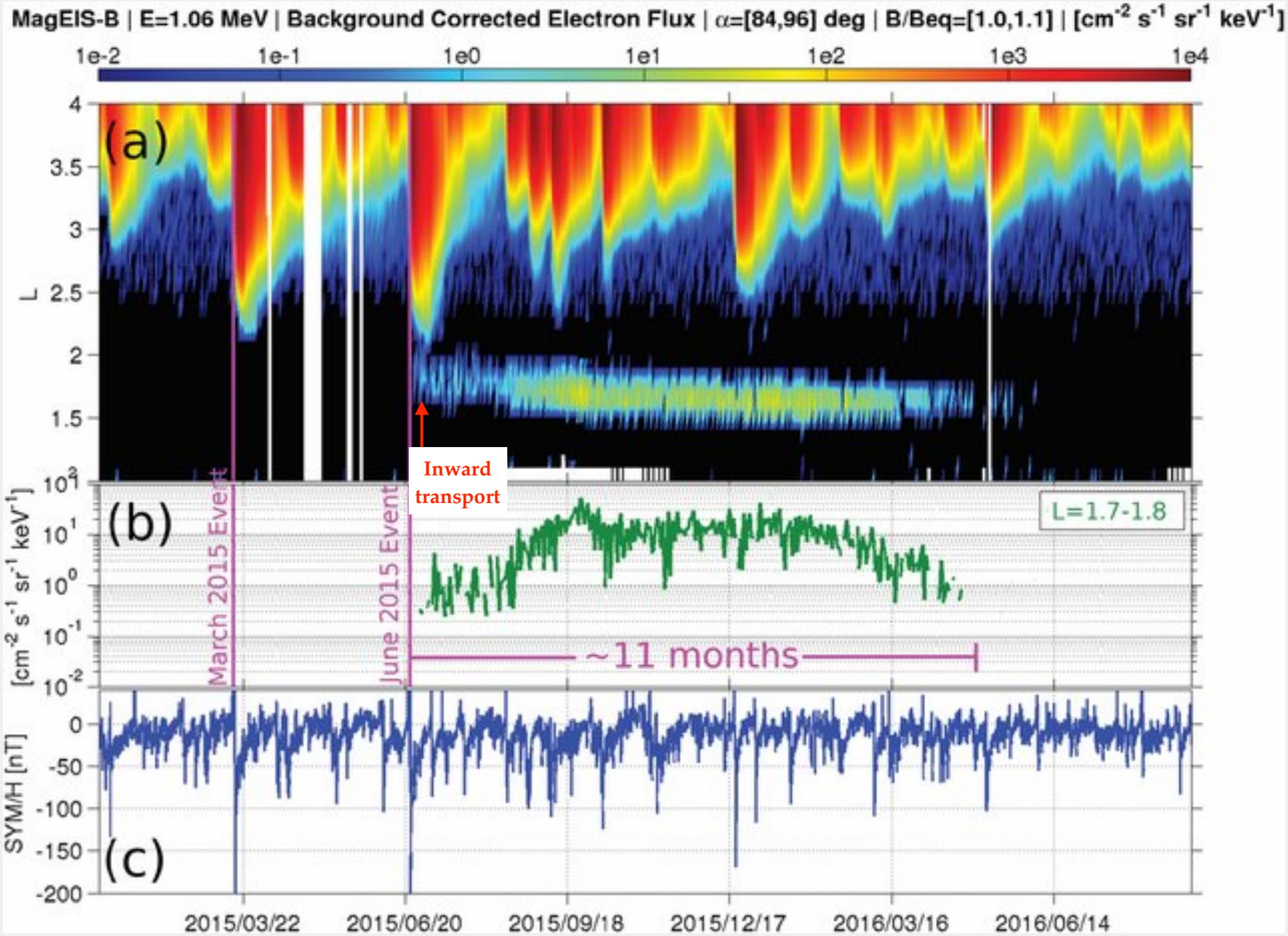
The Inner Belt (electrons): inward transport

SAMPEX Shows Traditional Two Belt Structure

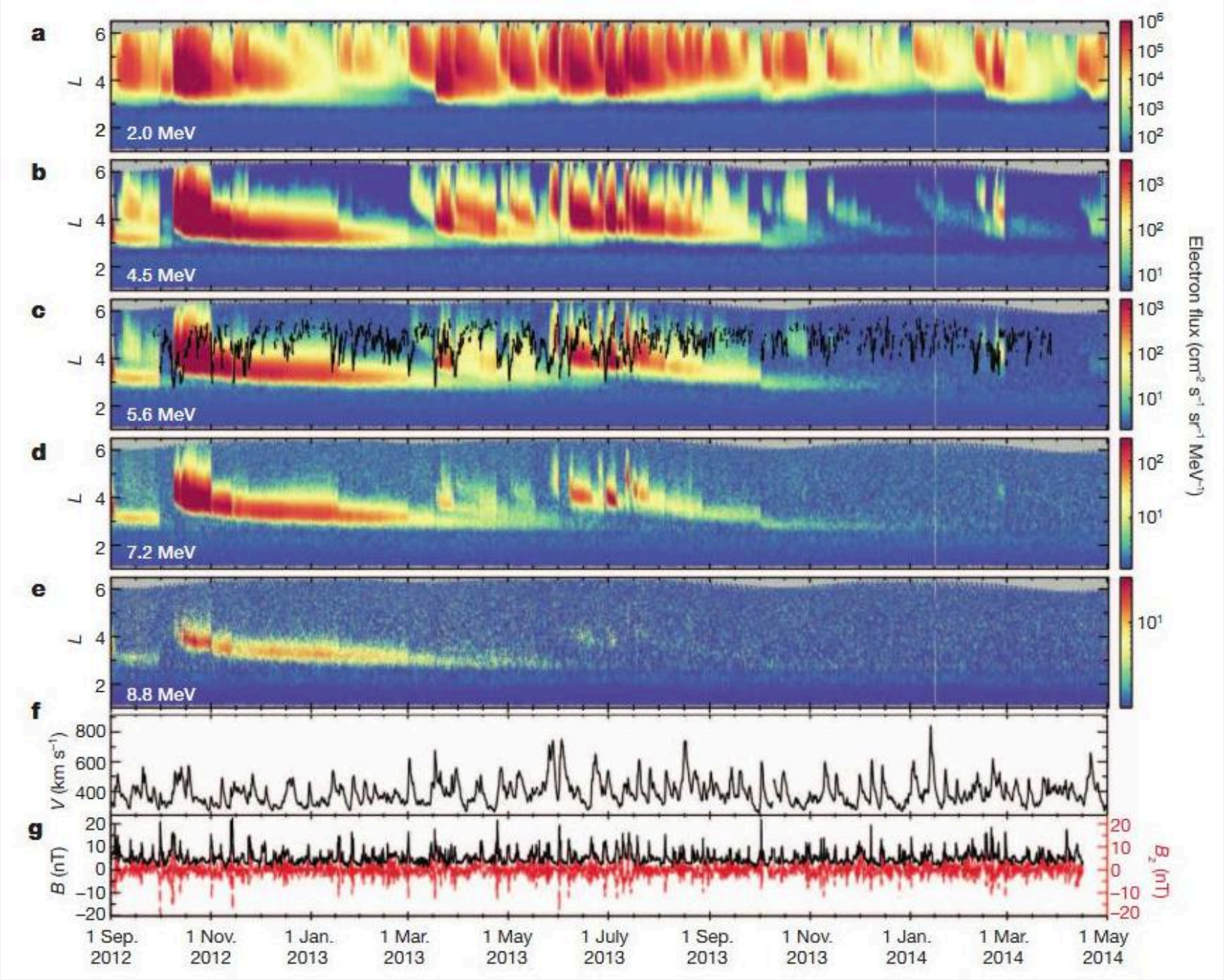
Long term (~12 year) plot from SAMPEX shows the established two belt structure



Inner Belt electrons: an example of enhancements

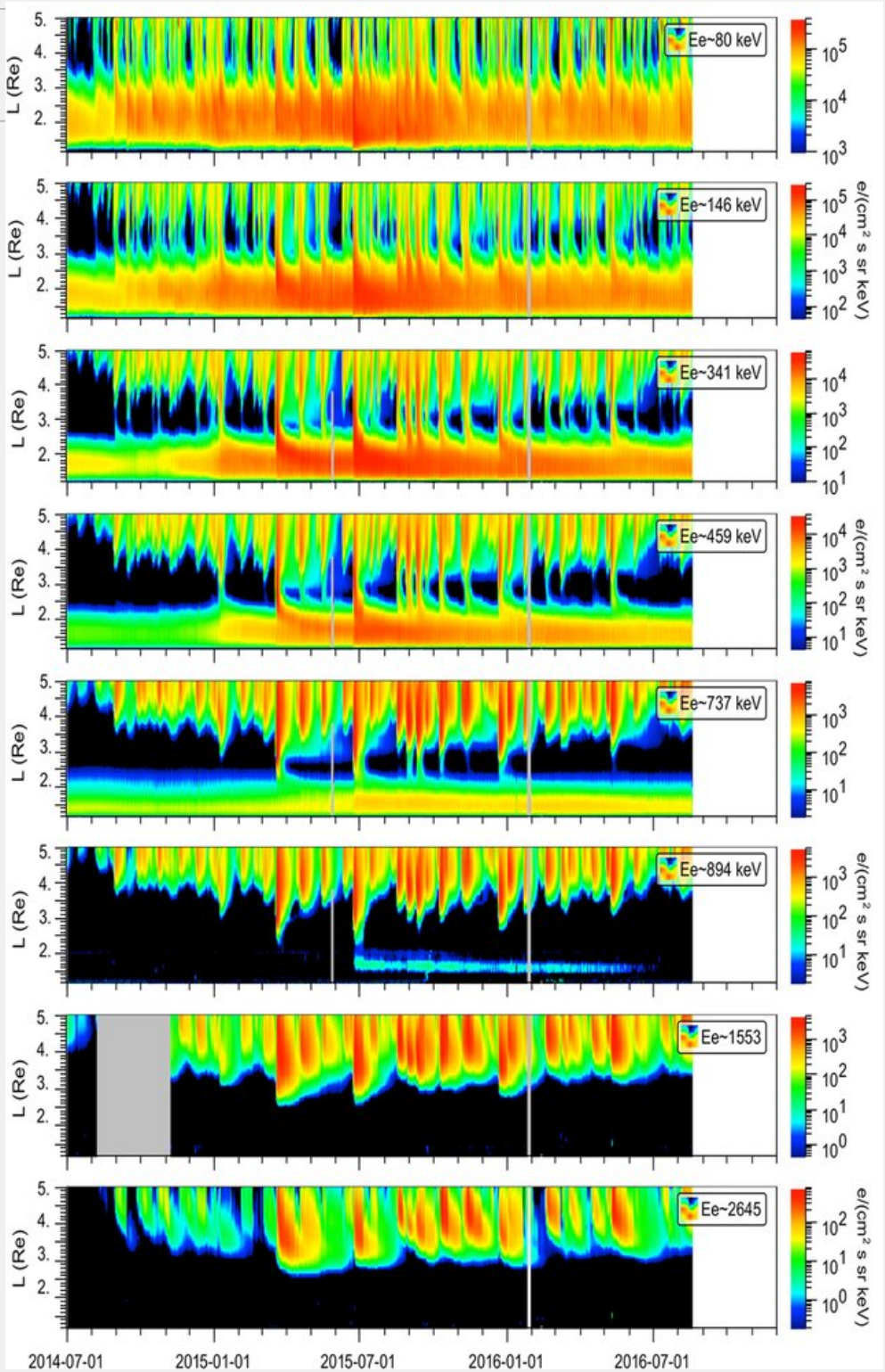


Variation at the highest energies



Comparison between Rad Belts and Ring Current

- Energies below ~300 keV: frequent injections and abundant in the inner L-shells (1.5 < L < 3, longer lifetimes there)
- This is the extension of the plasma sheet /ring current closer to the Earth
- “Medium” energies (400 keV – 800 keV) starting to form two-belt structure
- Higher energies (> 1 MeV) mostly confined to outer belt



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Radiation Effects

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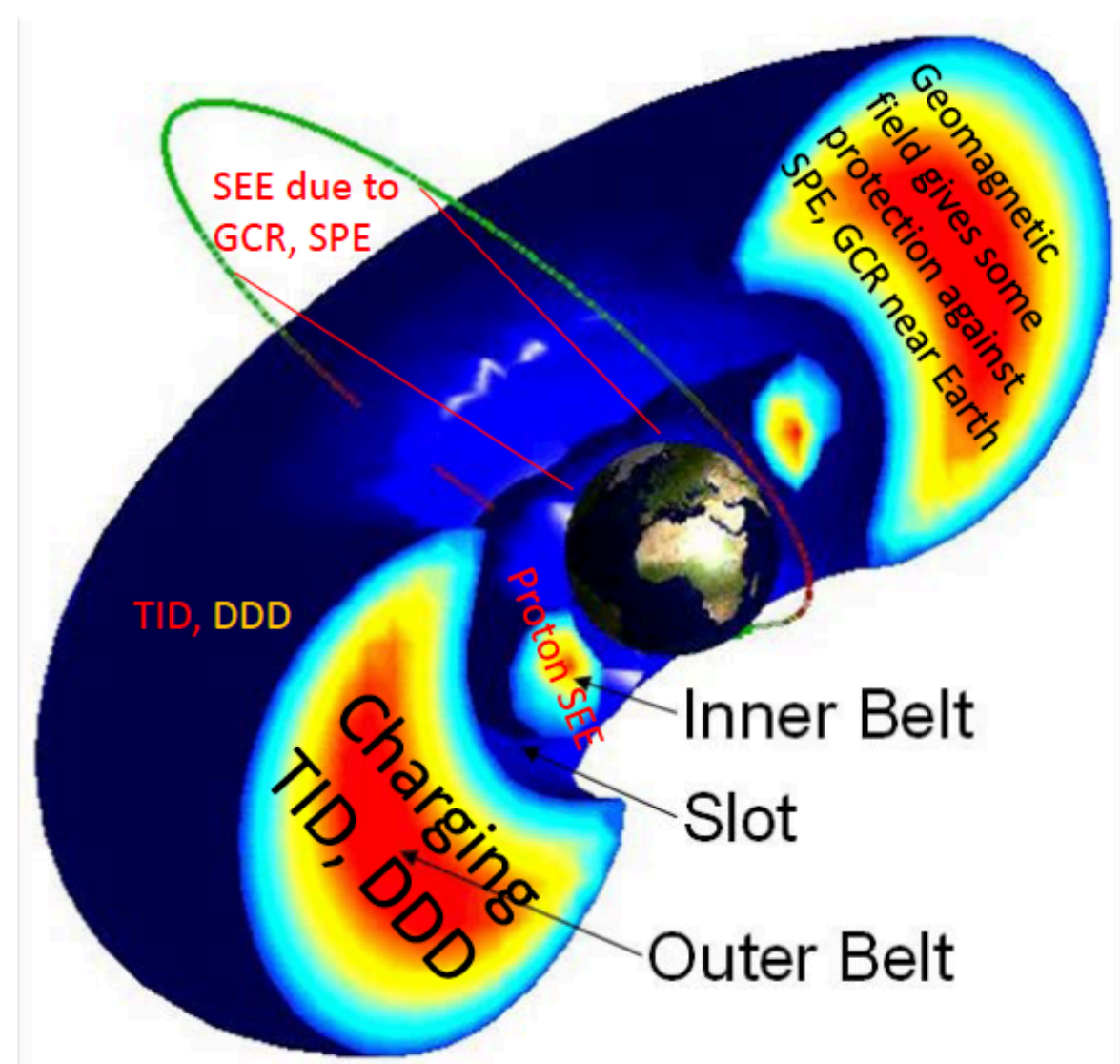
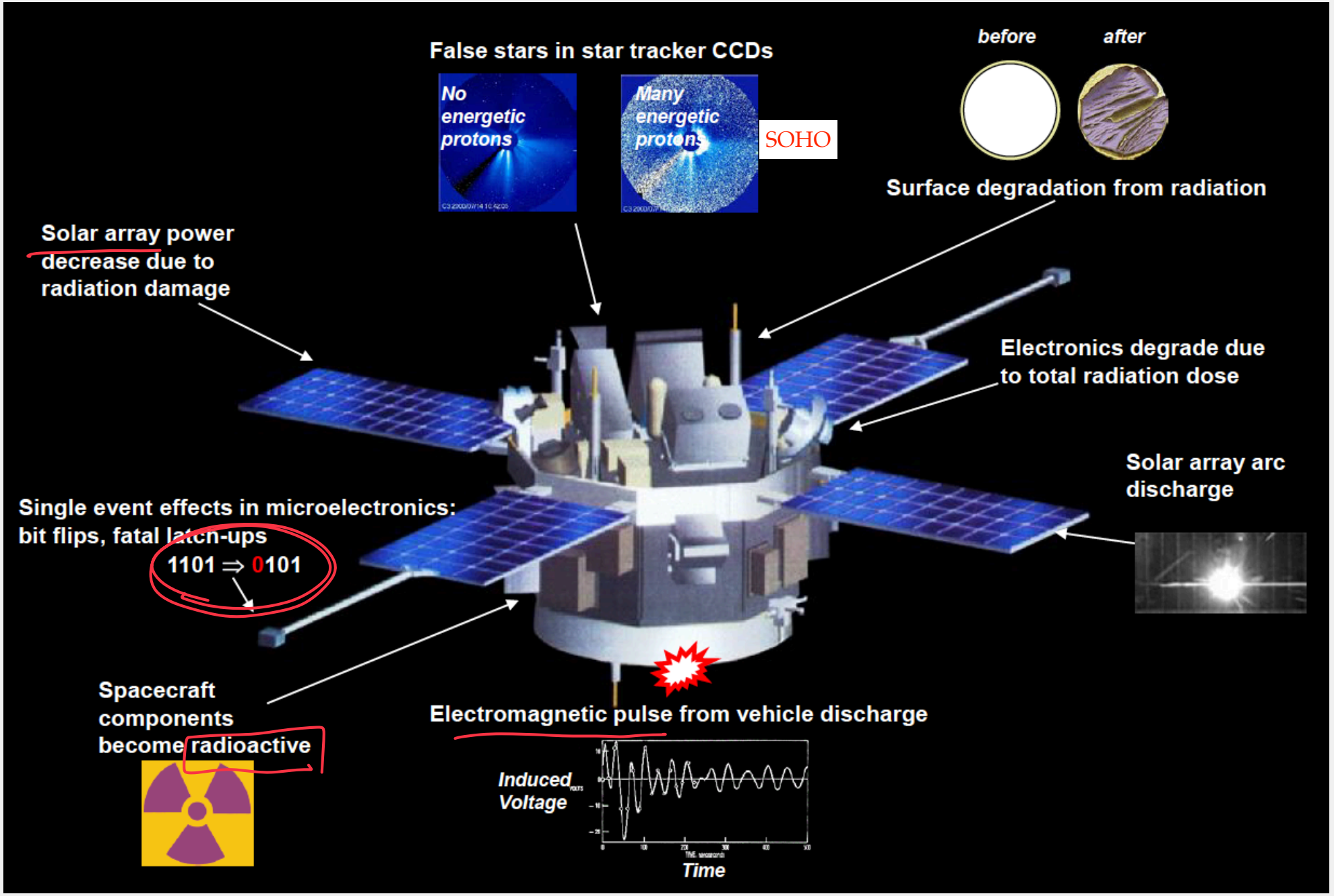
Radiation Effects on Spacecraft Electronics

- Radiation primarily affects spacecraft electronics
- **Total dose effects**: cumulative damage due to prolonged exposure
- **Deep dielectric discharge** (DDD)
- **Single event effects** (SEE)
- Solar array degradation; change in surface thermal characteristics



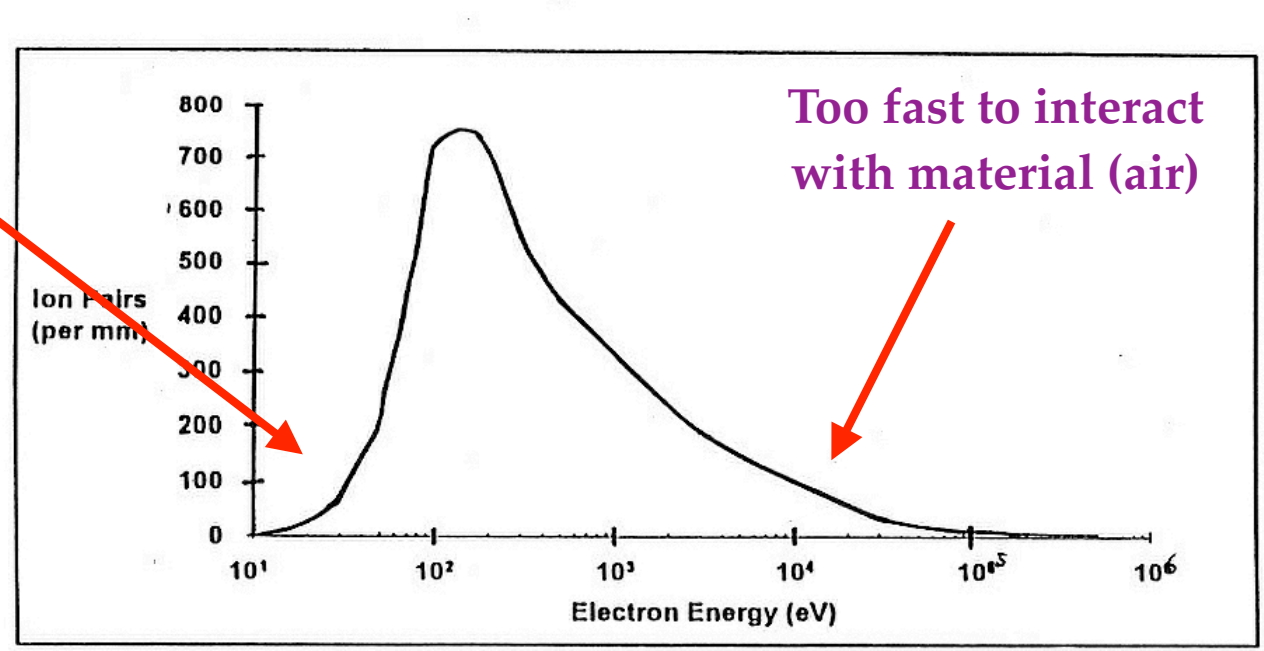
Radiation Effects



Ionizing Radiation

- We care about radiation because it causes ionization
  - Ionization causes damage if it accumulates too fast, or if accumulated charge becomes too high
  - Rate and effect of ionization depends on incident **particle type, particle energy**, and the **material**

Too slow to create ion pairs



The Ionization of air by an energetic electron.

Total Ionizing Dose

- Cumulative Effect; damage to electronics through prolonged exposure
- Electronic devices have some threshold, above which probability of failure increases dramatically
- Units: **rad** (radiation absorbed dose)
- **1 rad = 0.01 J/kg**: amount of energy absorbed per unit mass
- In SI units it has been replaced by the Gray (=1 J/kg), but most users in the US refuse to change
  - 1 Gray = 100 rad

Radiation Dose Tolerances

25 rad:	lowest dose to cause clinically observable blood changes
200 rad:	local dose for onset of erythema in humans
400 rad:	whole body LD_{50} for acute radiation syndrome in humans
1 krad:	whole body LD_{100} for acute radiation syndrome in humans[7]
1 krad:	typical radiation tolerance of ordinary microchips
4 to 8 krad:	typical radiotherapy dose, locally applied
10 krad:	fatal whole-body dose in 1964 Wood River Junction criticality accident[8]
1 Mrad:	typical tolerance of radiation-hardened microchips

Material	Damage Threshold (Rad)
Biological matter	$10^1 - 10^2$
Electronics	$10^2 - 10^6$
Lubricants, hydraulic fluid	$10^5 - 10^7$
Ceramics, glasses	$10^6 - 10^8$
Polymeric material	$10^7 - 10^9$
Structural metals	$10^9 - 10^{11}$

Other Radiation Units

- Radioactivity: amount of ionizing radiation released by a material. **Units: Curie (Ci:** 3.7x10^10 disintegration per second**)** or **Bequerel (Bq:** SI unit, 1 Ci= 3.7x10^10 Bq**)**
- Exposure: describes the amount of radiation traveling through the air. **Units: Roentgen** or **C/kg**
- Absorbed dose: **Units: rads** or **Grays**
- Dose equivalent: combines the amount of radiation absorbed and the medical effects of that type of radiation. Alpha particles have a greater effect than energy alone suggests. **Units: Sieverts (Sv)** or **rem** (roentgen equivalent in man)

For electrons,
1 R = 1 rad = 1 rem

exposure of 1 millirem from:

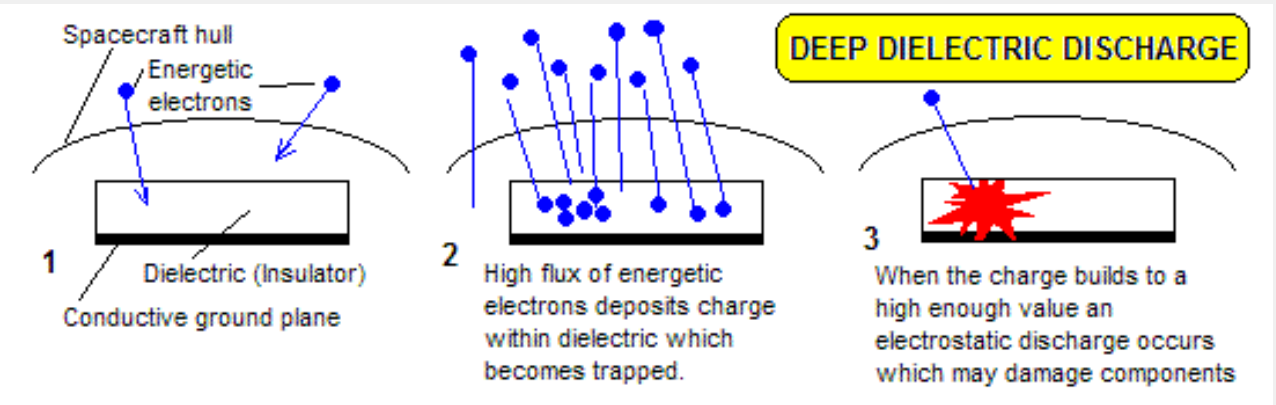
- 3 days of living in Atlanta
- 2 days of living in Denver
- 1 year of watching television (on average)
- 1 year of wearing a watch with a luminous dial
- 1 coast-to-coast airline flight
- 1 year living next door to a normally operating nuclear power plant

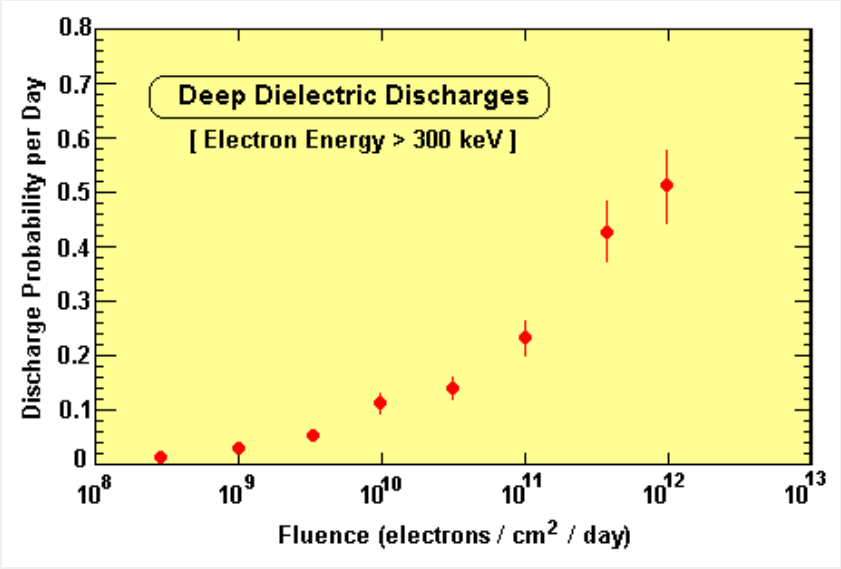
What is the dose rate in the heart of the radiation belts (in rads/yr) ?

- Say we had only 1 MeV electrons in the outer radiation belt, with a peak flux of 2×10^6 electrons/cm^2/sec.

Deep Dielectric Discharge

- High energy electrons penetrate the spacecraft outer surface and bury themselves in dielectric materials
  - circuit boards, cable wraps, feedthroughs, insulation around cables
- Buildup of charge will continue until the dielectric strength of the material is exceeded (typically in excess of 100,000 V/cm)
- Sudden electrical discharge will occur – "miniature lightning stroke"
- can cause permanent damage in the associated or nearby circuitry.

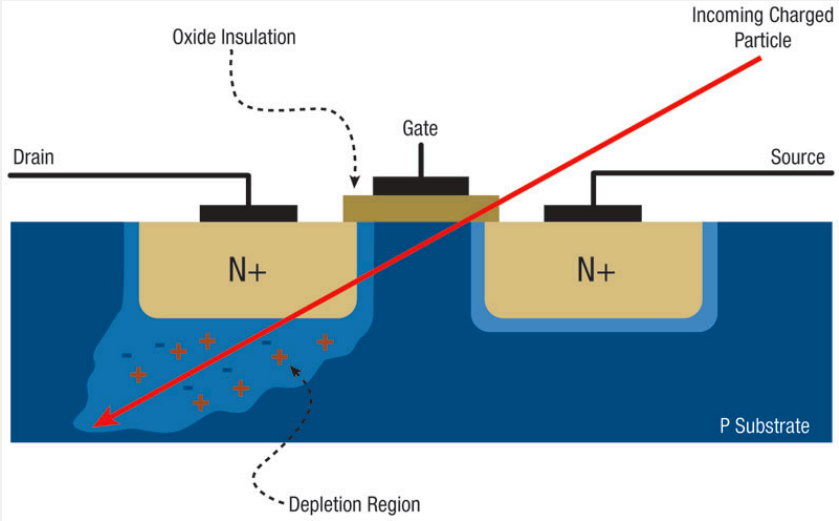


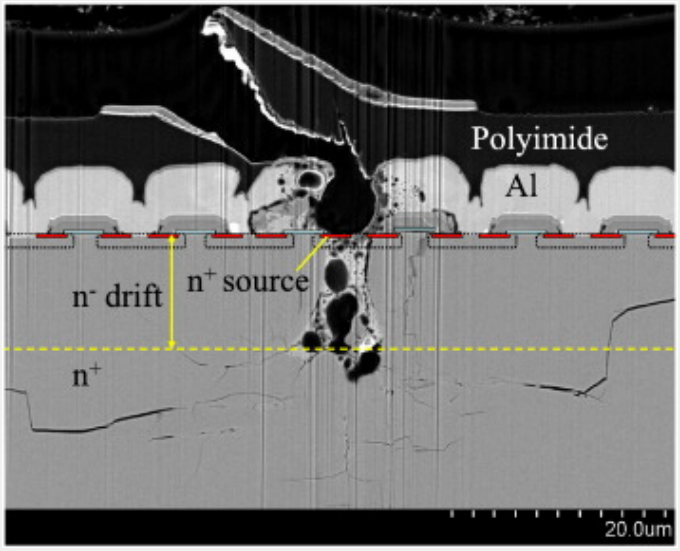


- Rules and models exist to predict DDD
- USAF satellite operators are issued warnings when the daily > 2 MeV electron fluence at GEO exceeds:
  - 3×10^8 el/cm^2/str/day for 3 days
  - 10^9 for a single day.

Single Event Effects

- Single Event: caused by an individual energetic particle
- **Single Event Upset (SEU)**: change of state in electronic system
  - common: flip a “0” to a “1”
  - Soft error: not damaging or persistent
- **Single Event Latchup (SEL)**: particle flips transistor into “on” state, drawing current
  - Requires power-cycle
  - May cause damage due to overcurrent
- **Single Event Burnout (SEB)**: turns on parasitic transistor in device; draws overcurrent and burns out device
- **Single Event Gate Rupture (SEGR):** a catastrophic failure in power metal-oxide-semiconductor field-effect transistors (MOSFETs) caused by a single, energetic particle, such as a heavy ion from cosmic radiation.





Other Radiation Effects

- Solar cell degradation
- Materials degradation: change in thermal properties (absorptivity, emissivity)
- Degradation of coatings, films, adhesives, etc.

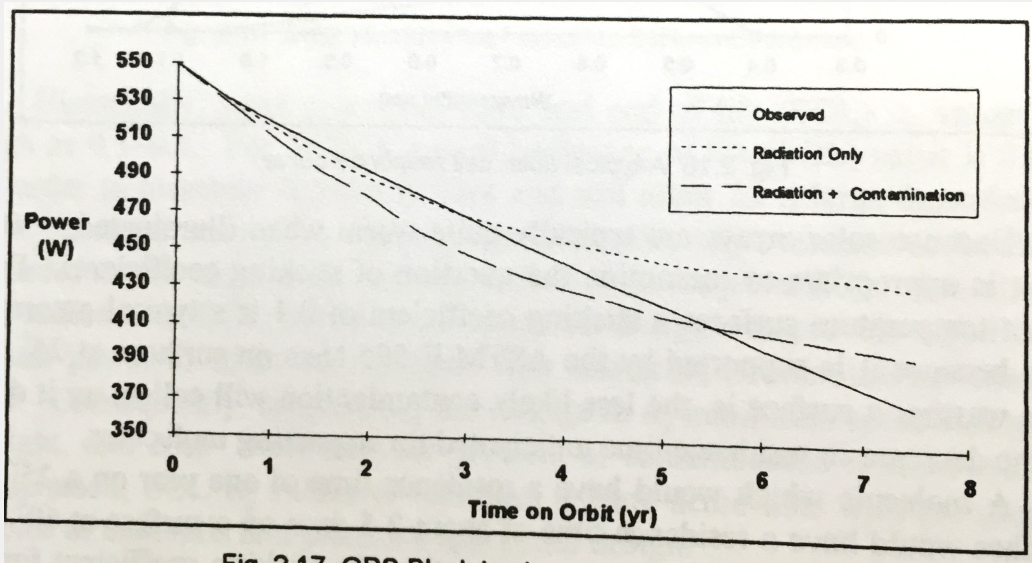
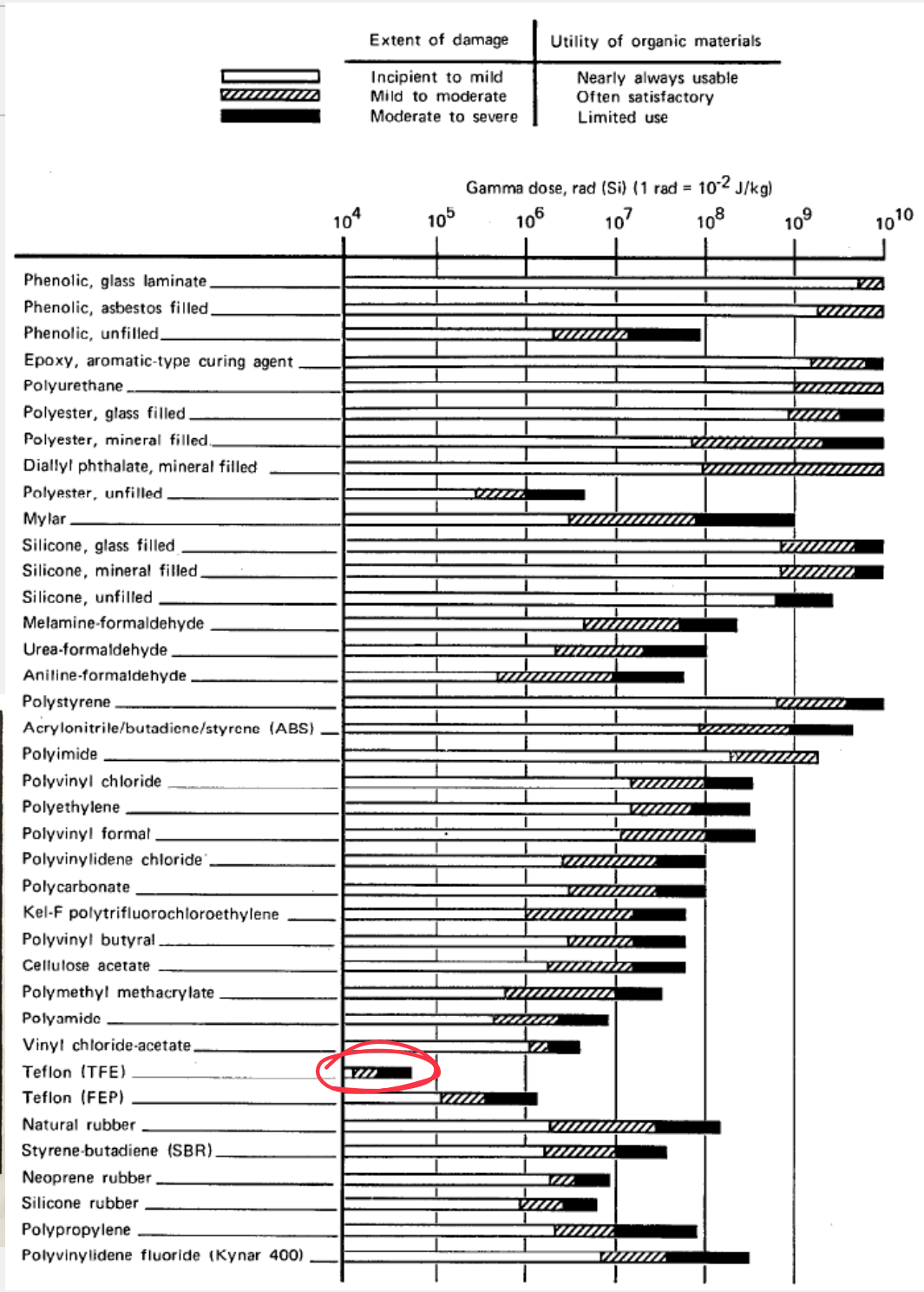


Fig. 2.17 GPS Block I solar power degradation.
(©1991 AIAA—reprinted with permission)



System Effects on Spacecraft

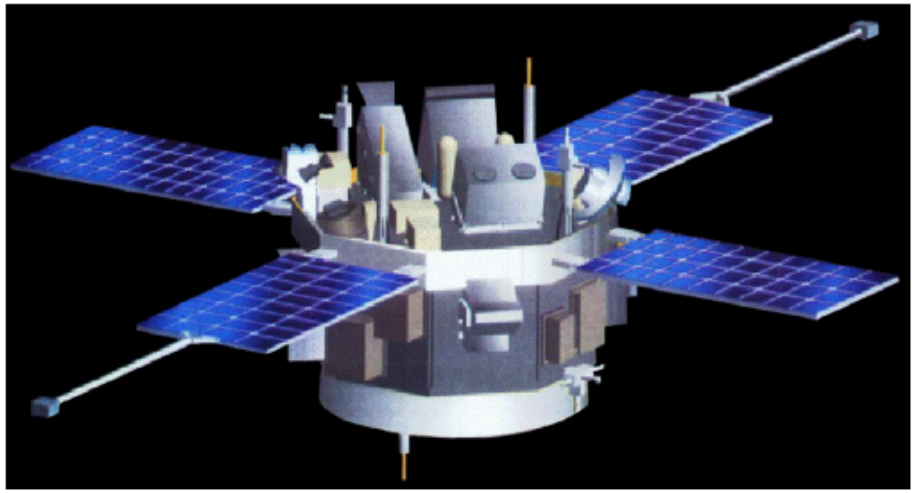
Effects of Space Radiation

Solar Panels
Degraded power from TID/DDD
ESD from surface charging

Guidance and Navigation
Station-keeping/pointing errors from SEE
Star tracker false stars (SEE) and noisy pixels (DDD)

Communications
Bit flips/lost data due to SEE/ESD

Avionics
DDD/TID degradation/failure
SEE/ESD malfunction/failure



Thermal from TID/plasma
Darkening of surfaces
Materials degradation

Power Electronics
DDD/TID degradation/failure
Destructive SEE failure

Optics
Darkening of optics from TID
Detector degradation from DDD

Contamination
Materials degraded by TID/DDD, atomic O, charging contaminate optics, etc.

Payload
Loss of data, availability, total loss due to SEE, ESD
Degradation, eventual failure due to TID, DDD

Surface/Materials
Degradation due to plasma, atomic oxygen, TID, charging

Different Particles and Energies have Different Effects in Different Regions

- electrons and protons < 100 keV: surface charging
- > 100 keV electrons: deep charging
- lower-energy protons: only TID
- higher-energy protons: single-event effects

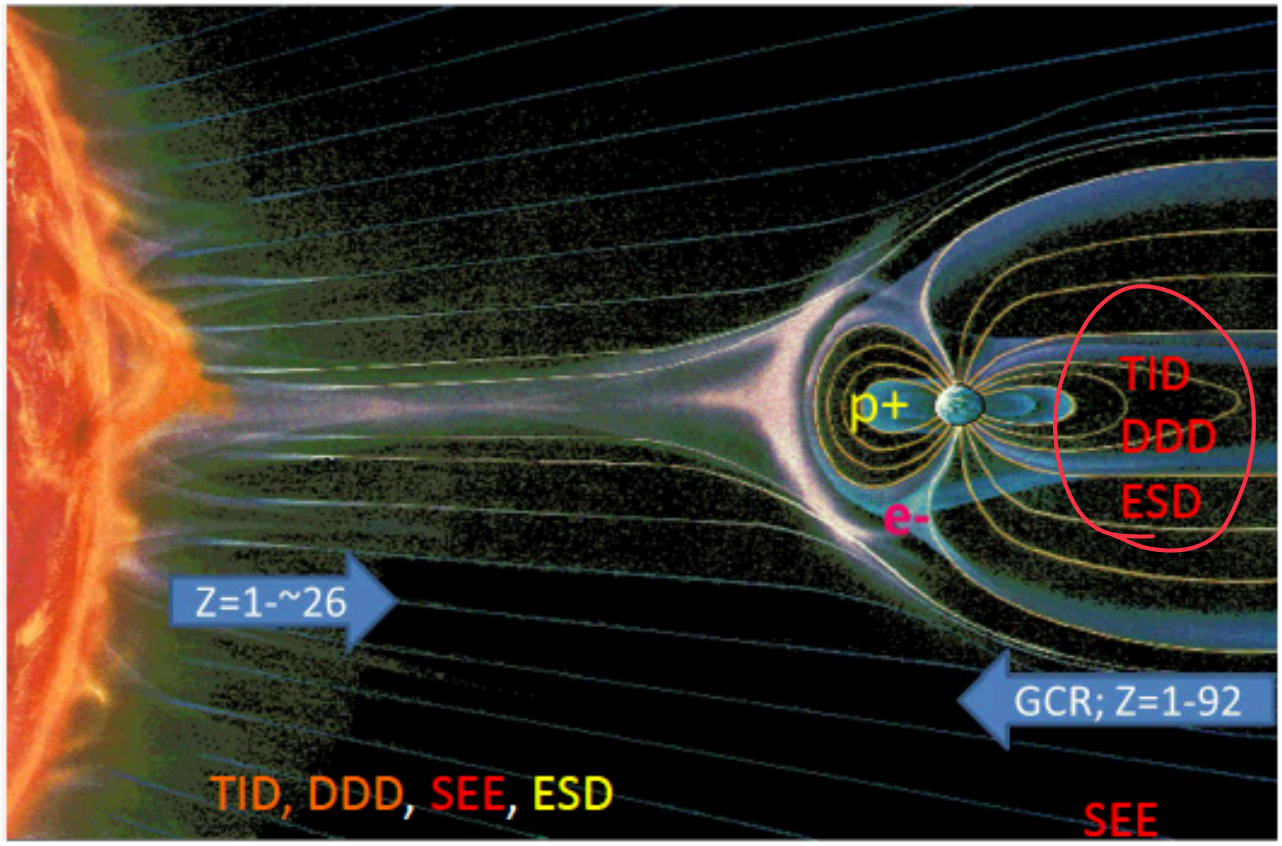
} All of these: TID

Priority	Species	Energy	Location	Sample Period	Effects
1	Protons	>10 MeV (> 80 MeV)	LEO & MEO	Mission	Dose, SEE, DD, nuclear activation
2	Electrons	> 1 MeV	LEO, MEO & GEO	5 min, 1 hr, 1 day, 1 week, & mission	Dose, internal charging
3	Plasma	30 eV – 100 keV (30 eV – 5 keV)	LEO, MEO & GEO	5 min, 1 hr, 1 day, 1 week, & mission	Surface charging & dose
4	Electrons	100 keV – 1 MeV	MEO & GEO	5 min, 1 hr, 1 day, 1 week, & mission	Internal charging, dose
5	Protons	1 MeV – 10 MeV (5 – 10 MeV)	LEO, MEO & GEO	Mission	Dose (e.g. solar cells)

Hot

Radiation Sources and Effects

- Exposed to different sources in different locations in space
- Effects vary based on energy, particle type

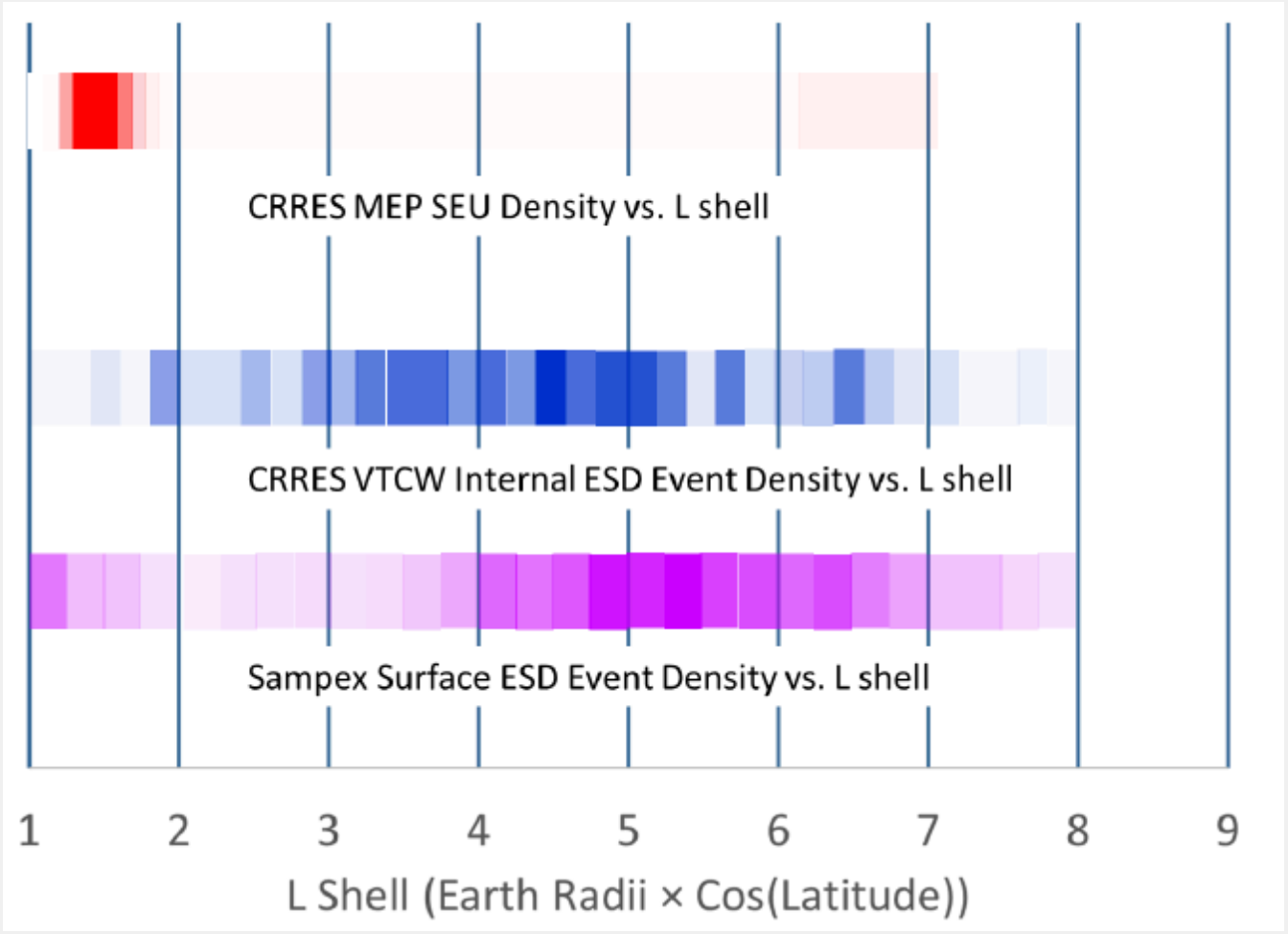


- **Galactic Cosmic Rays (GCR)—VERY high energy, All Z**
  - Small flux means main concern is SEE, even hard devices
- **Solar Particle Event (SPE)—Moderate energy, Z variable**
  - Intermittent high flux—SEE, TID, DDD all concerns
- **Trapped radiation belts—mainly protons and electrons**
  - High, variable flux; protons—TID,DDD, SEE; e^-— TID,DDD, charging (electrostatic discharge (ESD))

Different Effects in different regions

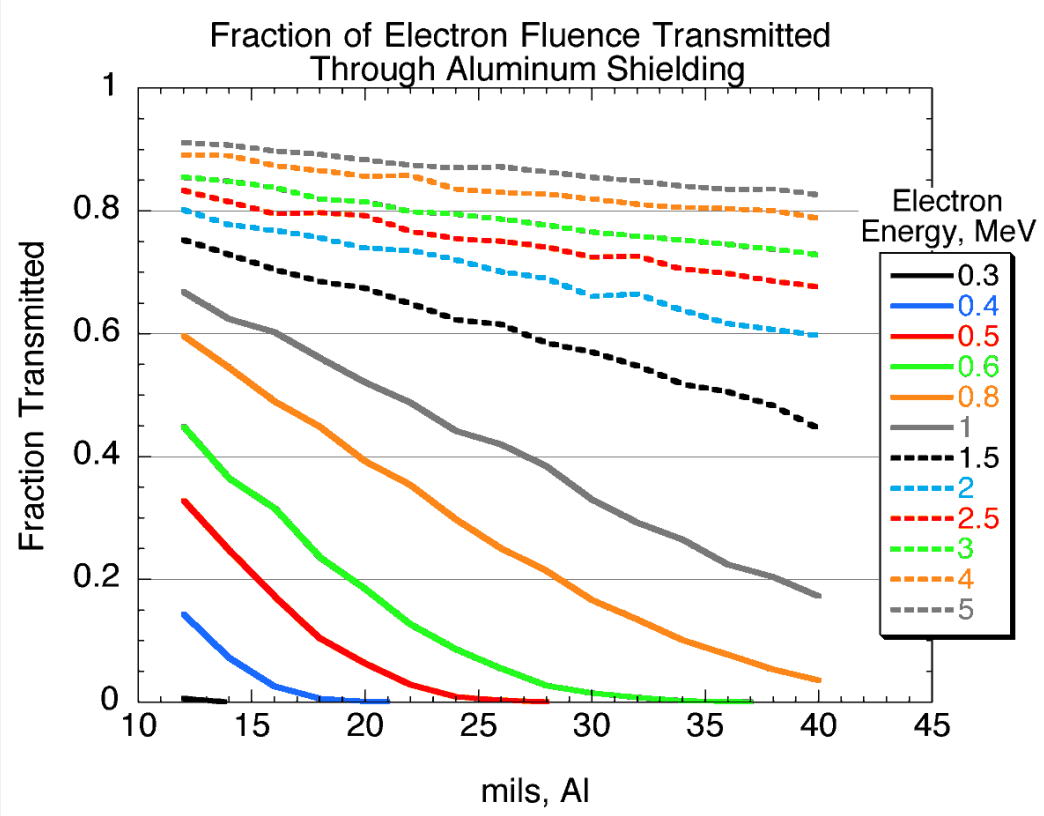
Attribution of Anomalies

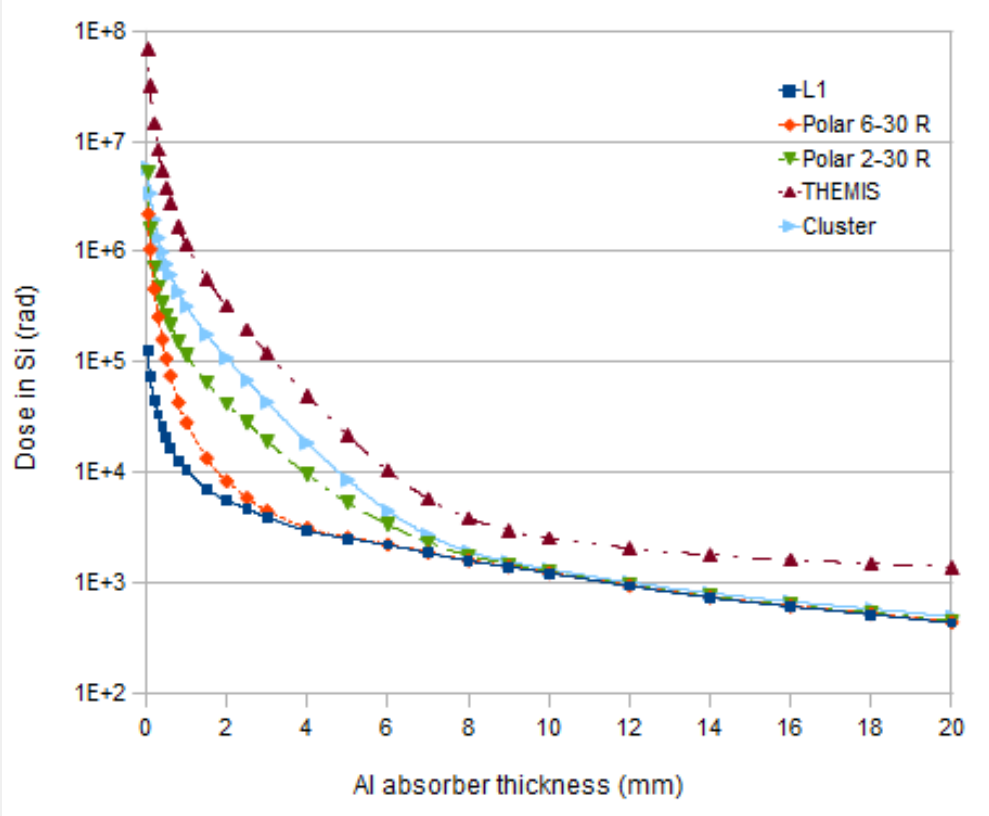
Diagnosis	Number of Forms
ESD - Internal Charging	74
ESD - Surface Charging	59
ESD - Uncategorized	28
Surface Charging	1
Total ESD & Charging	**162**
SEU - Cosmic Ray	15
SEU - Solar Particle Event	9
SEU - South Atlantic Anomaly	20
SEU - Uncategorized	41
Total SEU	**85**
Solar Array - Solar Proton Event	9
Total Radiation Dose	3
Materials Damage	3
South Atlantic Anomaly	1
Total Radiation Damage	**16**
Micrometeorid/Debris Impact	10
Solar Proton Event - Uncategorized	9
Magnetic Field Variability	5
Plasma Effects	4
Atomic Oxygen Erosion	1
Atmospheric Drag	1
Sunlight	1
IR background	1
Ionospheric Scintillation	1
Energetic Electrons	1
Other	2
Total Miscellaneous	**36**



Mitigation of Radiation Effects

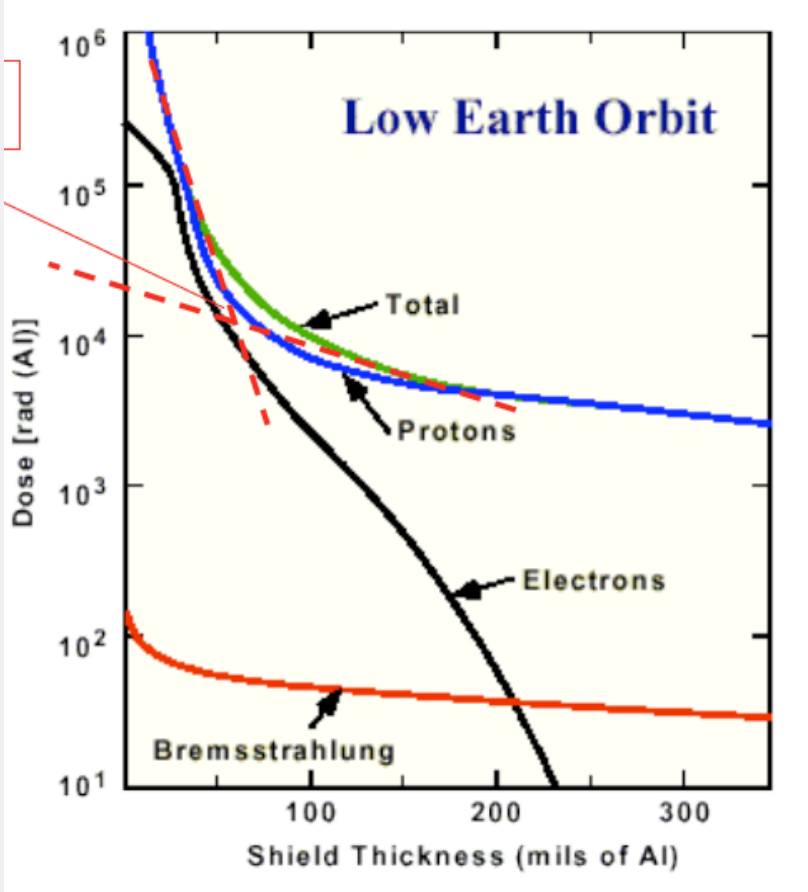
- Common techniques: i) Shielding; ii) redundancy / voting; iii) radiation hardening
- Shielding reduces exposure to ionizing radiation and therefore dose, extending life
- Also reduces probability of SEE events



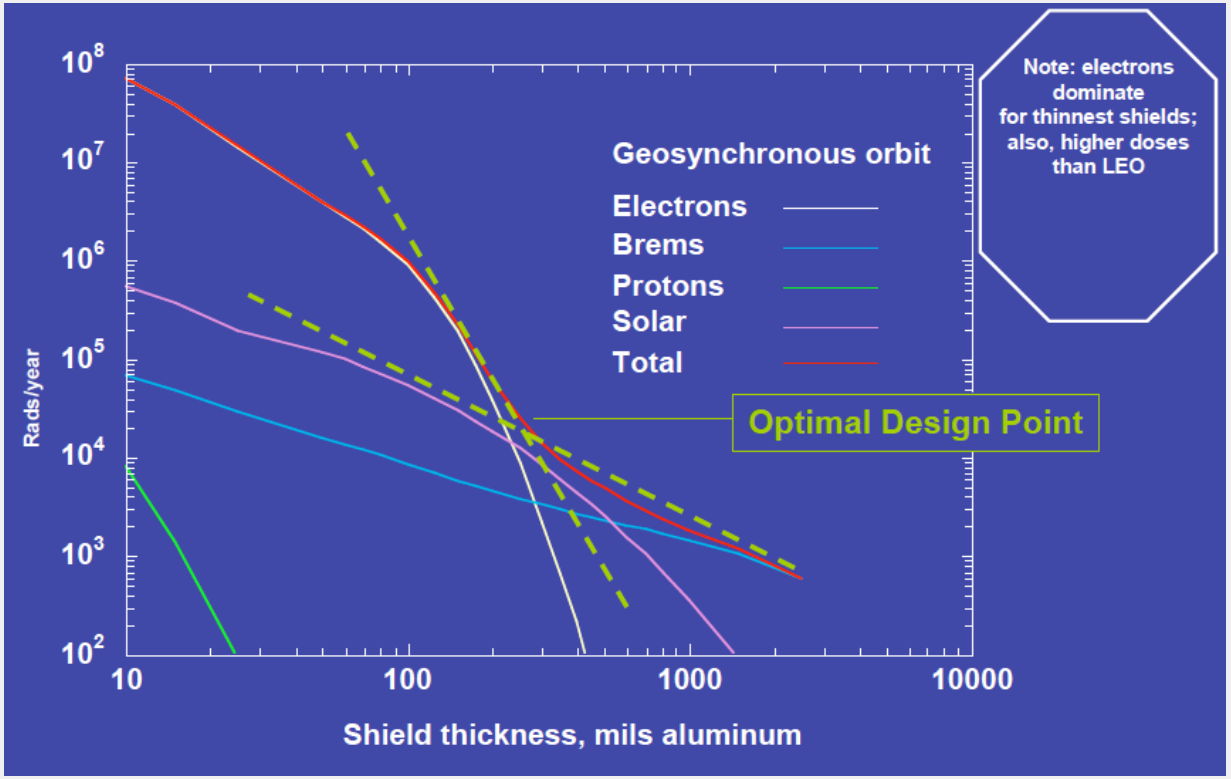


https://www.spenvis.oma.be/

Shielding in LEO and GEO



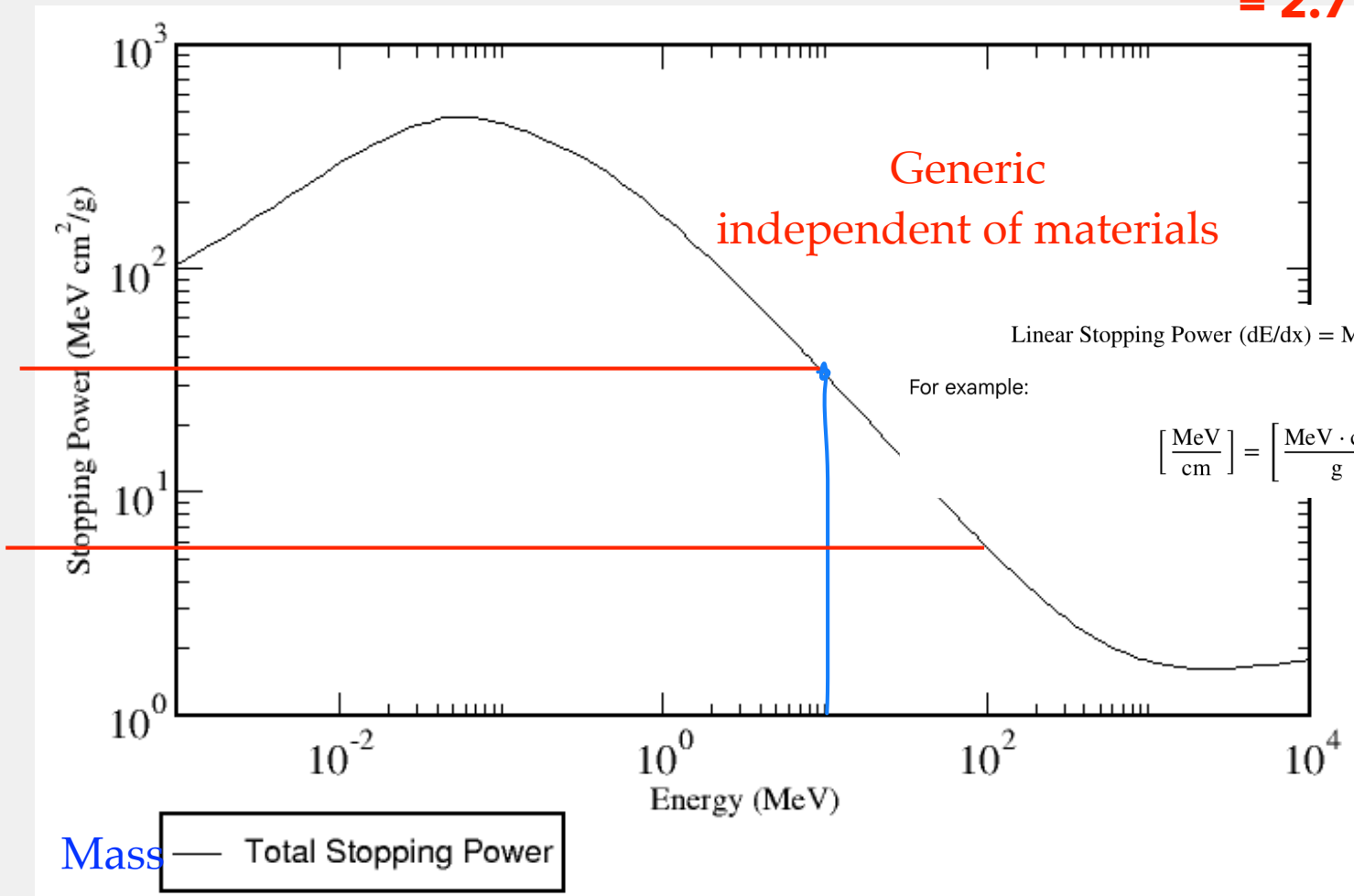
- LEO has considerably lower dose than GEO
- LEO dominated by protons; GEO dominated by electrons, up to a point
- "Optimal design point" where increased thickness of Aluminum has diminishing returns



Mass stopping power, the actual stopping power and range: protons

- The Actual (Linear) Stopping Power (dE/dx) of a specific material: multiply the curve by its density
- For Aluminum, at 10 MeV, ~36 MeV cm^2/g x 2.7 g/cm^3 = 97.2 MeV/cm (actual stopping power, dE/dx, for 10 MeV protons)
- **That means 10 MeV protons will be stopped in d = 0.1 cm (Range, shielding thickness)**
- Same calculation for 100 MeV gives d = 6.7 cm (100./(5.5*2.7)=6.734 cm)

Aluminum density = 2.7 g/cm^3



Linear Stopping Power (dE/dx) = Mass Stopping Power × Density

For example:

$$\left[\frac{\text{MeV}}{\text{cm}}\right] = \left[\frac{\text{MeV}\cdot\text{cm}^2}{\text{g}}\right] \times \left[\frac{\text{g}}{\text{cm}^3}\right]$$

Mass stopping power, actual stopping power and range: electrons

- For Aluminum, a 1 MeV, ~1.5 MeV cm^2/g x 2.7 g/cm^3 = 4 MeV/cm (actual stopping power, dE/dx)
- **That means 1 MeV electrons will be stopped in d = 0.25 cm (Range, shielding thickness)**
- Same calculation for 10 MeV gives d = 2 cm (~1.9*2.7 =5.13 MeV/cm)

Aluminum density = 2.7 g/cm^3